INCH-POUND

MIL-M-28787/509
5 MAY 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BUS ARBITRATOR MODULE, KEY CODE BWD

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic bus arbitrator module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issues of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 2.415 failures per million operating hours for class I or 5.080 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be BWD.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B15	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1-D6	Ibd	V	S	-0.027	0.122	-0.027	0.122	-0.027	0.122	mA
D7-D11	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D12	Ibd	V	S	-0.05	2.04	-0.05	2.04	-0.05	2.04	mA
D13	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D14	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D15	Ibd	V	S	-0.027	0.122	-0.027	0.122	-0.027	0.122	mA
D16	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D17,D18	Ibd	V	S	-0.027	0.122	-0.027	0.122	-0.027	0.122	mA
D19	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D20	Ibd	V	S	-0.027	0.122	-0.027	0.122	-0.027	0.122	mA
D21	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D22	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D23	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D24	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D25	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D26	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D27	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D28,D29	Ibd	V	S	-0.027	0.122	-0.027	0.122	-0.027	0.122	mA
D30	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D31	Ibd	V	S	-0.027	0.122	-0.027	0.122	-0.027	0.122	mA
D32	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D33-D35	Ibd	V	S	-0.027	0.122	-0.027	0.122	-0.027	0.122	mA
E1-E5	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E6-E11	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	mA
E12	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E13	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E14-E20	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	mA
E21	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 8 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E22	Ios	VI	S	-120.0	-20.0	-120.0	-20.0	-120.0	-20.0	nA
E23	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	nA
E24	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	nA
E25,E26	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	nA
E27	Ios	VI	S	-326.0	-101.0	-326.0	-101.0	-326.0	-101.0	nA
E28	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	nA
E29,E30	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	nA
E31-E36	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	nA
E37	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	nA
E38	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	nA
E39	Ios	VI	S	-326.0	-101.0	-326.0	-101.0	-326.0	-101.0	nA
E40-E46	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	nA
F1-F6	Iih	VII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	nA
F7-F11	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	nA
F12	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	nA
F13	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	nA
F14	Iih	VII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	nA
F15	Iih	VII	X	-0.9	0.04	-0.9	0.04	-0.9	0.04	nA
F16,F17	Iih	VII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	nA
F18	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	nA
F19	Iih	VII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	nA
F20	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	nA
F21	Iih	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	nA
F22	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	nA
F23	Iih	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	nA
F24	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	nA
F25	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	nA
F26	Iih	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	nA
F27	Iih	VII	X	-2.91	-0.454	-2.91	-0.454	-2.91	-0.454	nA
F28,F29	Iih	VII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	nA
F30	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	nA
F31	Iih	VII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	nA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F32	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F33-F35	Iih	VII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	mA
F36	Iih	VII	X	-4.52	-1.02	-4.52	-1.02	-4.52	-1.02	mA
G1-G6	Iil	VIII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	mA
G7-G11	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G12	Iil	VIII	X	-4.52	-1.02	-4.52	-1.02	-4.52	-1.02	mA
G13	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G14	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G15	Iil	VIII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	mA
G16	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G17,G18	Iil	VIII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	mA
G19	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G20	Iil	VIII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	mA
G21	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G22	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G23	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G24	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G25	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G26	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G27	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G28	Iil	VIII	X	-2.91	-0.454	-2.91	-0.454	-2.91	-0.454	mA
G29,G30	Iil	VIII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	mA
G31	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G32	Iil	VIII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	mA
G33	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G34-G36	Iil	VIII	X	-1.66	-0.544	-1.66	-0.544	-1.66	-0.544	mA
H1-H48	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J45	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1-L7	Vlc	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L8	Vlc	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L9-L18	Vlc	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
M1	PSFC	XII	S	4.2	8.0	4.2	8.0	4.2	8.0	uF
M2	PSFC	XII	S	5.0	9.4	5.0	9.4	5.0	9.4	uF
N	Icc	XIII	X	830	2000	830	2000	830	2000	mA
P1,P2	Functional	XIV	X	See table XIV						
Q1,Q2	Dynamic	XV	S	See table XV						
Q3,Q4	Dynamic	XV	Q	See table XV						
R1-R5	C1/o	XVII	Q	4	100	4	100	4	100	pF
R6	C1/o	XVII	Q	4	75	4	75	4	75	pF
R7-R10	C1/o	XVII	Q	2	50	2	50	2	50	pF
R11	C1/o	XVII	Q	2	125	2	125	2	125	pF
R12	C1/o	XVII	Q	6	75	6	75	6	75	pF
R13,R14	C1/o	XVII	Q	2	50	2	50	2	50	pF
R15	C1/o	XVII	Q	4	50	4	50	4	50	pF
R16	C1/o	XVII	Q	4	75	4	75	4	75	pF
R17	C1/o	XVII	Q	4	100	4	100	4	100	pF
R18	C1/o	XVII	Q	4	125	4	125	4	125	pF
R19	C1/o	XVII	Q	2	75	2	75	2	75	pF
R20	C1/o	XVII	Q	4	75	4	75	4	75	pF
R21	C1/o	XVII	Q	2	50	2	50	2	50	pF
R22	C1/o	XVII	Q	2	100	2	100	2	100	pF
R23	C1/o	XVII	Q	2	50	2	50	2	50	pF
R24	C1/o	XVII	Q	4	75	4	75	4	75	pF
R25	C1/o	XVII	Q	2	50	2	50	2	50	pF
R26	C1/o	XVII	Q	2	75	2	75	2	75	pF
R27-R29	C1/o	XVII	Q	4	75	4	75	4	75	pF
R30	C1/o	XVII	Q	2	50	2	50	2	50	pF
R31	C1/o	XVII	Q	4	75	4	75	4	75	pF
R32	C1/o	XVII	Q	2	50	2	50	2	50	pF
R33	C1/o	XVII	Q	4	100	4	100	4	100	pF
R34	C1/o	XVII	Q	4	75	4	75	4	75	pF
R35	C1/o	XVII	Q	4	100	4	100	4	100	pF
R36-R39	C1/o	XVII	Q	4	75	4	75	4	75	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 80 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R40	C1/o	XVII	Q	10	150	10	150	10	150	pF
R41-R55	C1/o	XVII	Q	2	50	2	50	2	50	pF
R56	C1/o	XVII	Q	4	75	4	75	4	75	pF
R57	C1/o	XVII	Q	2	100	2	100	2	100	pF
R58	C1/o	XVII	Q	4	175	4	175	4	175	pF
R59	C1/o	XVII	Q	2	50	2	50	2	50	pF
R60	C1/o	XVII	Q	4	150	4	150	4	150	pF
R61	C1/o	XVII	Q	4	75	4	75	4	75	pF
R62	C1/o	XVII	Q	2	75	2	75	2	75	pF
R63-R68	C1/o	XVII	Q	2	50	2	50	2	50	pF
R69	C1/o	XVII	Q	4	75	4	75	4	75	pF
R70-R72	C1/o	XVII	Q	2	50	2	50	2	50	pF
R73-R75	C1/o	XVII	Q	4	75	4	75	4	75	pF
R76	C1/o	XVII	Q	8	100	8	100	8	100	pF
R77-R79	C1/o	XVII	Q	4	75	4	75	4	75	pF
R80	C1/o	XVII	Q	6	125	6	125	6	125	pF
T1, T2	PSTR	XVIII	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/509-1 for class I, and M28787/509-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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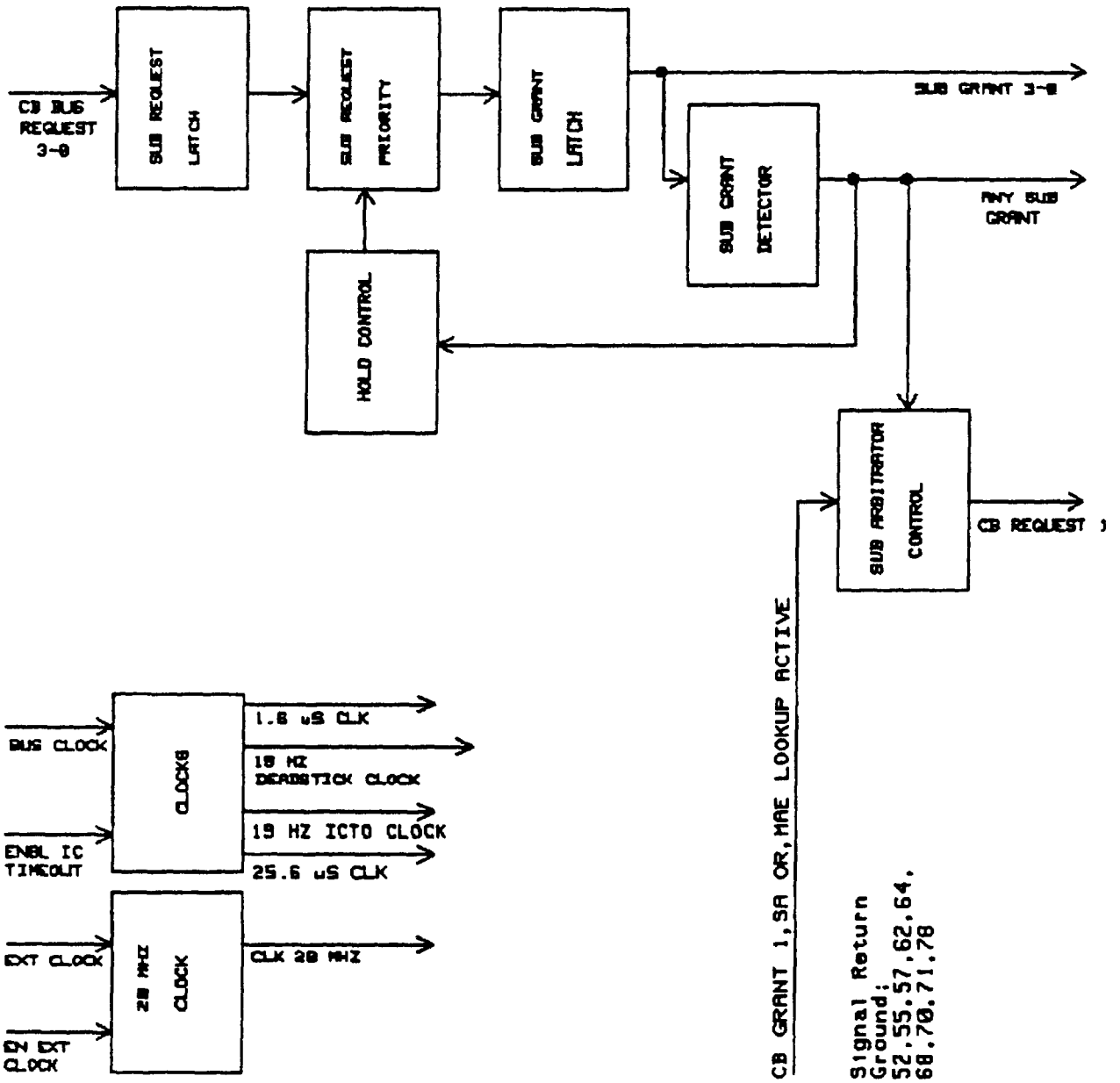


FIGURE 1. BWD functional block diagram.

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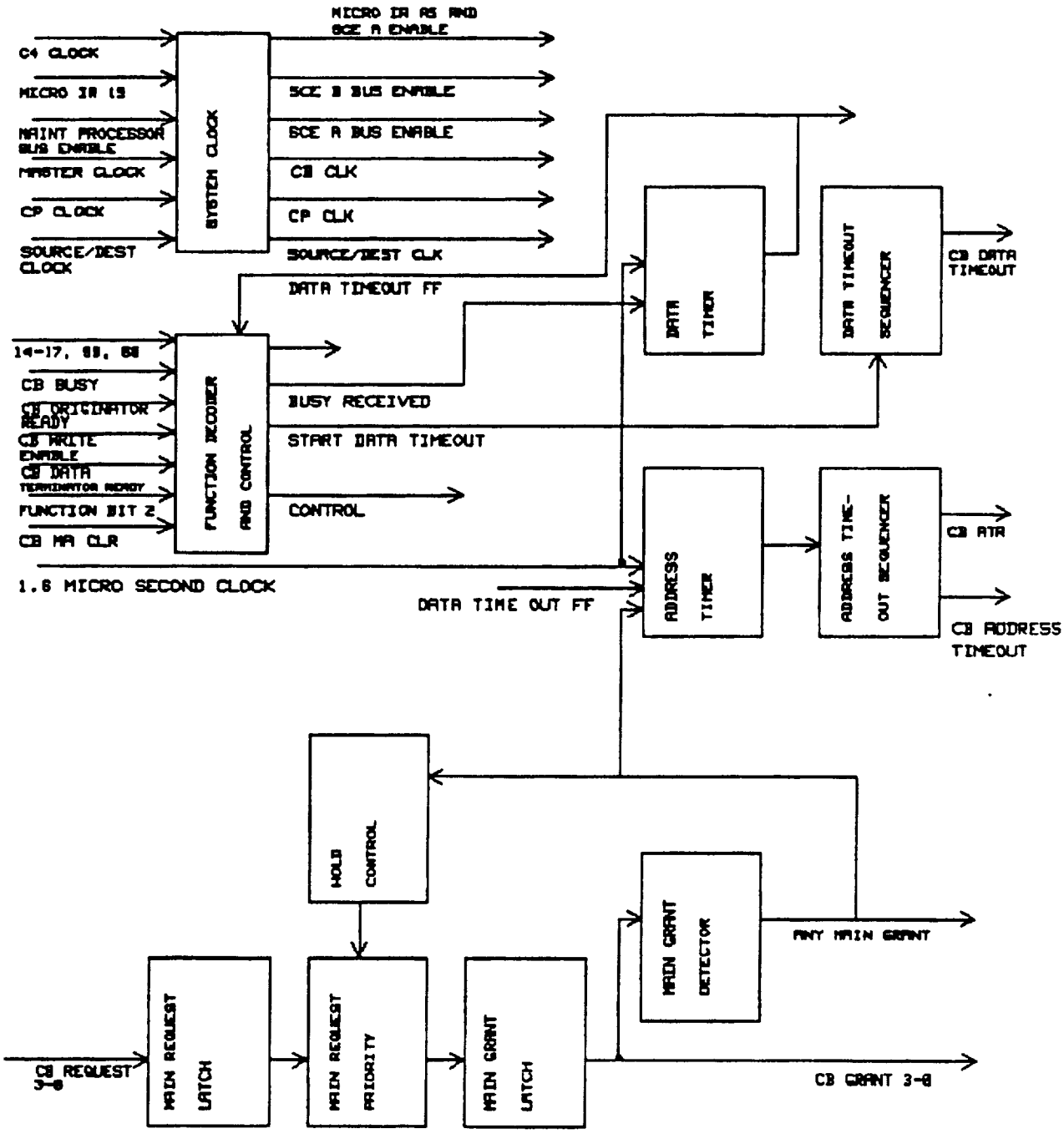


FIGURE 1. BWD functional block diagram - Continued.

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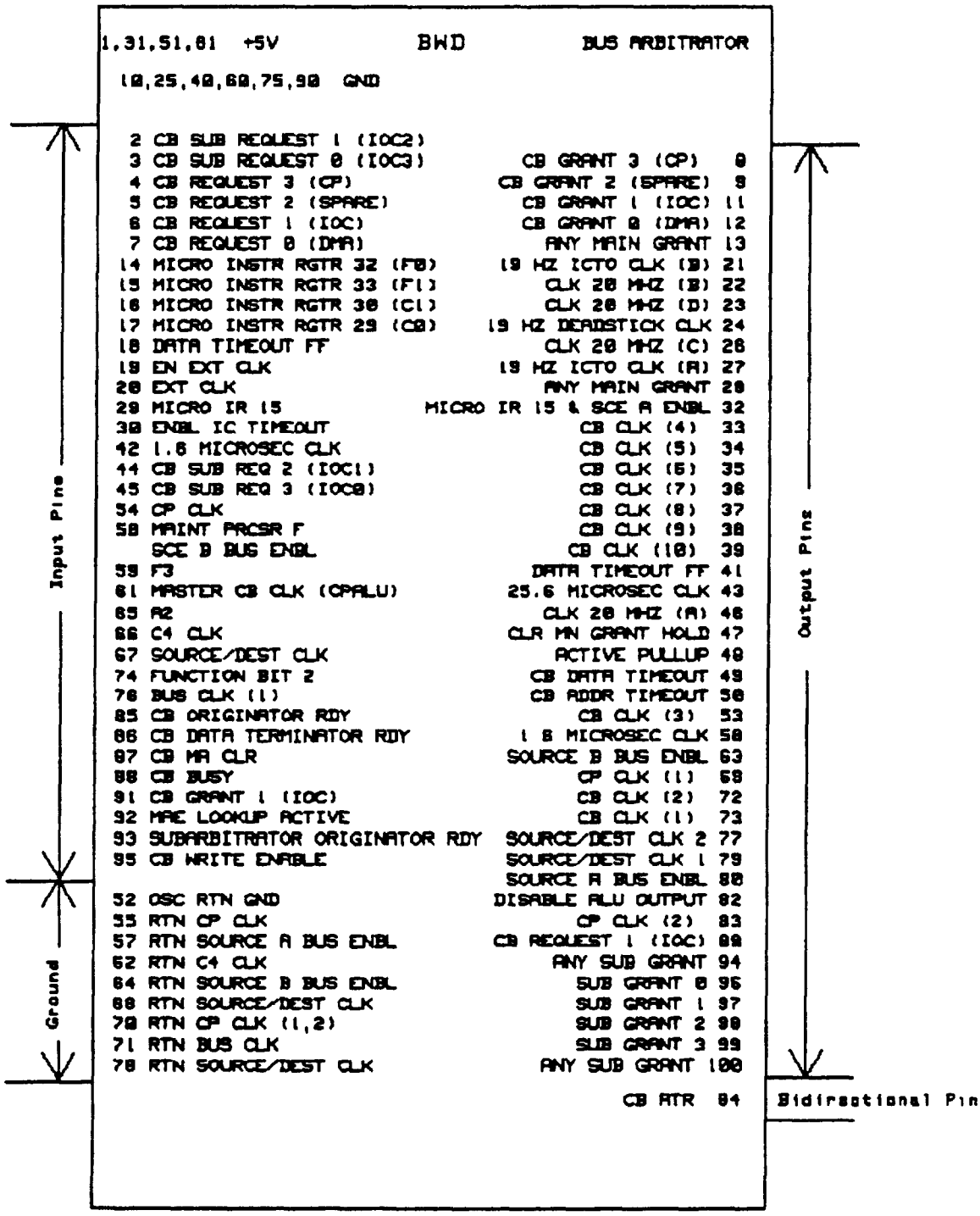


FIGURE 2. BWD pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 $\pm$ 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical

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testing is called for at the end of each temperature stabilization period, test conditions B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

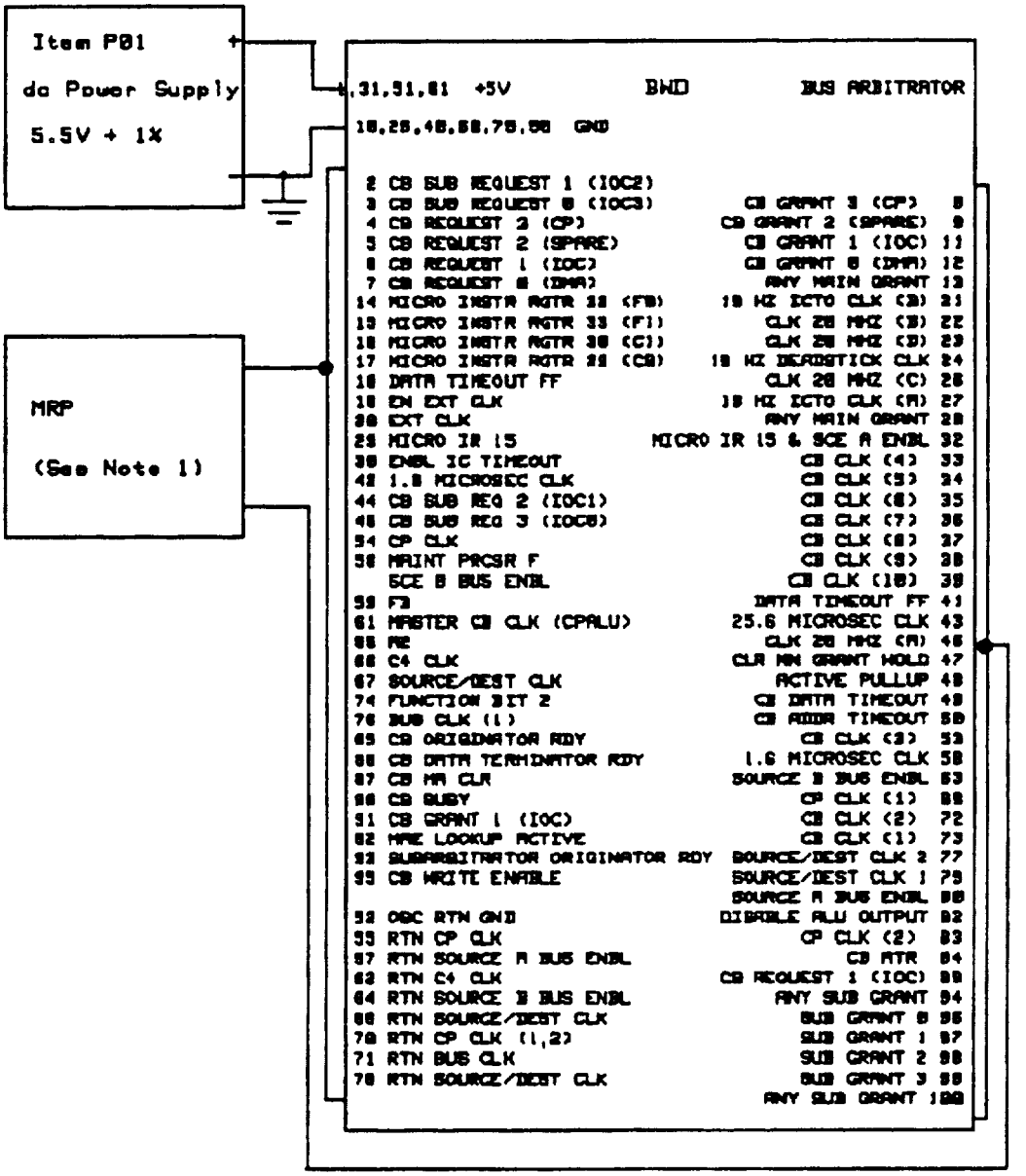
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. BWD operating temperature and component temperature test circuit.

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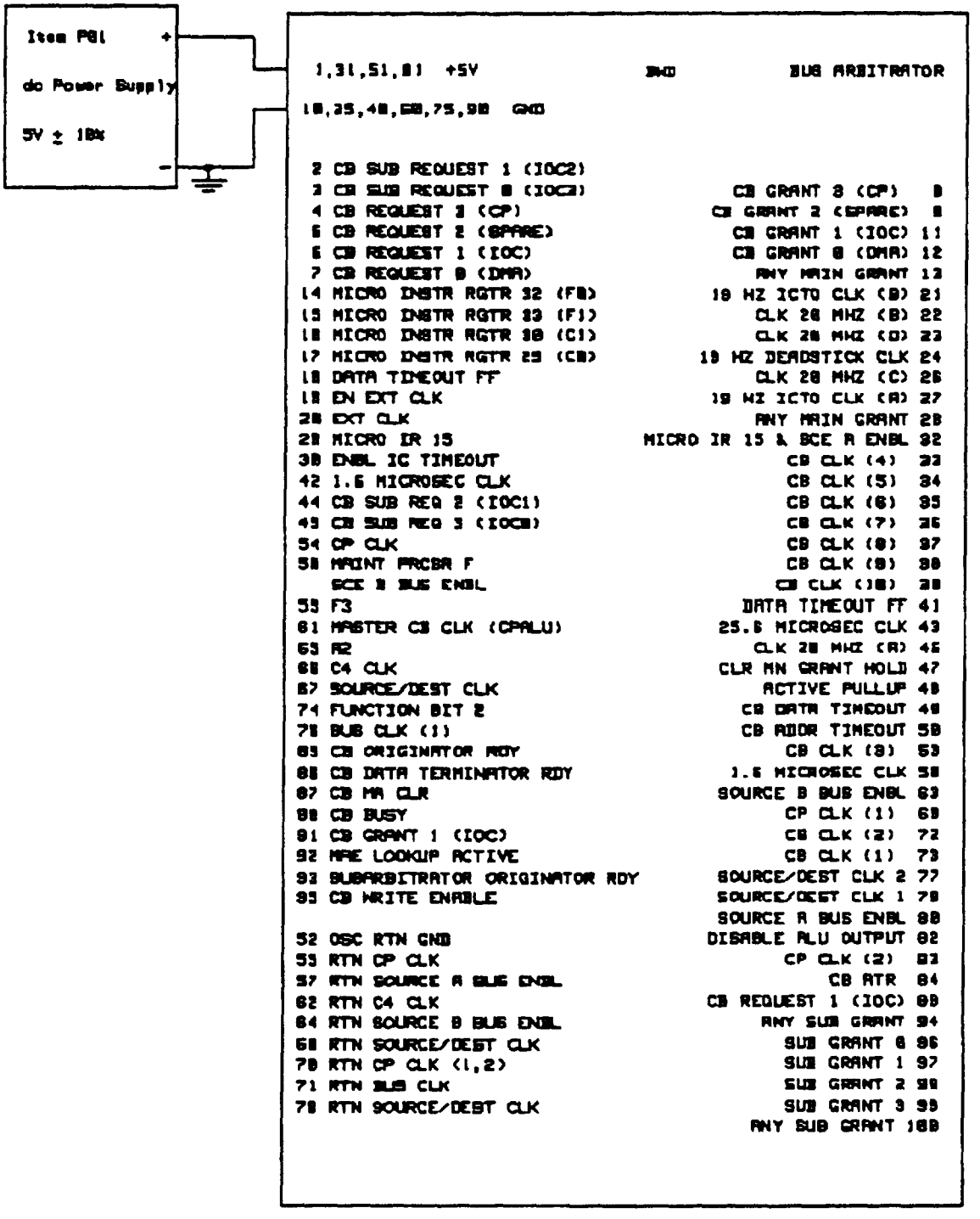
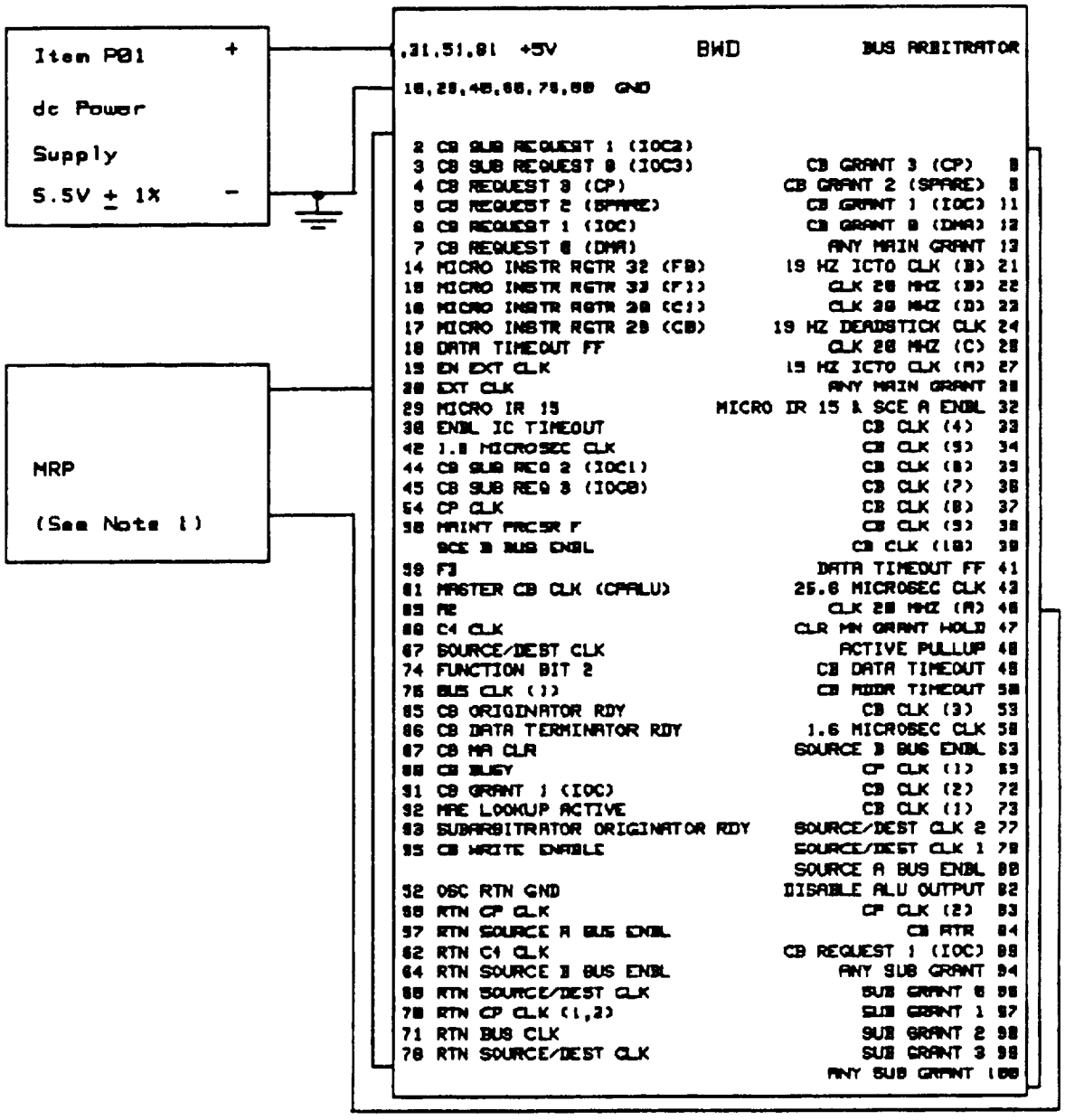


FIGURE 4. BWD vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. BWD life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input and output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vic = input clamp diode voltage.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Vin = input forcing voltage.
(s) Voh = output high level voltage.
(t) Vol = output low level voltage.
(u) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 11.0 W (maximum), 7.34 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB SUB REQ 1	NA	0.4	2.4	-1.66	-0.544	0.122	100
CB SUB REQ 0	NA	0.4	2.4	-1.66	-0.544	0.122	100
CB REQUEST 3	NA	0.4	2.4	-1.66	-0.544	0.122	100
CB REQUEST 2	NA	0.4	2.4	-1.66	-0.544	0.122	100
CB REQUEST 1	NA	0.4	2.4	-1.66	-0.544	0.122	100
CB REQUEST 0	NA	0.4	2.4	-1.66	-0.544	0.122	75
INS REG 32	-1.2	0.4	2.4	-0.6	0.02	0.1	50
INS REG 33	-1.2	0.4	2.4	-0.6	0.02	0.1	50
INS REG 30	-1.2	0.4	2.4	-0.6	0.02	0.1	50
INS REG 29	-1.2	0.4	2.4	-0.6	0.02	0.1	50
DTA TMOUT FF	-1.2	0.4	2.4	-0.6	0.02	0.1	125
EN EXT CLK	NA	0.4	2.4	-4.52	-1.02	2.04	75
EXT CLK	-1.2	0.4	2.4	-2.0	0.05	1.0	50
MICRO IR 15	-1.2	0.4	2.4	-0.6	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
EN IC TMOUT	NA	0.4	2.4	-1.66	-0.544	0.122	50
1.6 uS CLK	-1.5	0.4	2.4	-0.8	0.04	0.2	75
CB SUB REQ 2	NA	0.4	2.4	-1.66	-0.544	0.122	125
CB SUB REQ 3	NA	0.4	2.4	-1.66	-0.544	0.122	125
CP CLK	-1.2	0.4	2.4	-2.0	0.05	1.0	75
MAINT PRCSR	NA	0.4	2.4	-1.66	-0.544	0.122	75
F3	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MASTR CB CK	-1.2	0.4	2.4	-4.0	0.1	2.0	100
A2	-1.2	0.4	2.4	-0.6	0.02	0.1	50
C4 CLK	-1.2	0.4	2.4	-4.0	0.1	2.0	75
SRC/DEST CLK	-1.2	0.4	2.4	-2.0	0.05	1.0	50
FUNCT BIT 2	-1.2	0.4	2.4	-0.6	0.02	0.1	75
BUS CLK (1)	-1.2	0.4	2.4	-4.0	0.1	2.0	75
CB ORIG RDY	NA	0.4	2.4	-1.66	-0.544	0.122	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DTA TERM RDY	NA	0.4	2.4	-1.66	-0.544	0.122	75
CB MA CLR	-1.2	0.4	2.4	-2.0	0.05	1.0	50
CB BUSY	NA	0.4	2.4	-1.66	-0.544	0.122	75
CB GRANT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MAE LKUP ACT	NA	0.4	2.4	-1.66	-0.544	0.122	100
SUB ORG RDY	NA	0.4	2.4	-1.66	-0.544	0.122	75
CB WR EN	NA	0.4	2.4	-1.66	-0.544	0.122	100
CB ATR	NA	0.4	2.4	-2.91	-0.454	NA	NA

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
CB GRANT 3	2.4	0.5	0.4	6.0	-60.0	75
CB GRANT 2	2.4	0.5	0.4	6.0	-60.0	75
CB GRANT 1	2.4	0.5	0.4	6.0	-60.0	75
CB GRANT 0	2.4	0.5	0.4	6.0	-60.0	75
ANY MAIN GNT	2.4	0.5	0.4	6.0	-60.0	150
19HZ ICT B	2.4	0.5	0.4	NA	-50.0	50
CLK 20 MHZ B	2.4	0.5	0.4	6.0	-50.0	50
CLK 20 MHZ D	2.4	0.5	0.4	6.0	-50.0	50
19HZ DSTK CK	2.4	0.5	0.4	NA	-50.0	50
CLK 20 MHZ C	2.4	0.5	0.4	6.0	-50.0	50
19HZ ICT A	2.4	0.5	0.4	NA	-50.0	50
ANY MAIN GNT	2.4	0.5	0.4	6.0	-40.0	50
IR15 A ENBL	2.4	0.5	0.4	6.0	-60.0	50
CB CLK (4)	2.4	0.5	0.4	6.0	-50.0	50

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
CB CLK (5)	2.4	0.5	NA	6.0	-50.0	50
CB CLK (6)	2.4	0.5	0.4	6.0	-50.0	50
CB CLK (7)	2.4	0.5	0.4	6.0	-50.0	50
CB CLK (8)	2.4	0.5	0.4	6.0	-50.0	50
CB CLK (9)	2.4	0.5	0.4	6.0	-50.0	50
CB CLK (10)	2.4	0.5	0.4	6.0	-50.0	50
DTA TMOUT FF	2.4	0.5	0.4	6.0	-60.0	NA
25.6 uS CLK	2.4	0.5	-0.16	1.6	-20.0	75
CLK 20 MHZ A	2.4	0.5	-0.4	6.0	-50.0	100
CLR MN GNT H	2.4	0.5	-0.4	6.0	-60.0	175
ACTIV PULLUP	2.4	NA	-0.4	NA	-100.0	50
CB DTA TMOUT	2.4	0.5	-0.4	6.0	-100.0	150
CB ADR TMOUT	2.4	0.5	-0.4	6.0	-101.0	75
CB CLK (3)	2.4	0.5	-0.4	6.0	-50.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
1.6 uS CLK	2.4	0.5	-0.4	6.0	-40.0	50
SRC B BUS EN	2.4	0.5	-0.4	6.0	-40.0	50
CP CLK (1)	2.4	0.5	-0.4	6.0	-50.0	50
CB CLK (2)	2.4	0.5	-0.4	6.0	-50.0	50
CB CLK (1)	2.4	0.5	-0.4	6.0	-50.0	50
SRC/DEST CK2	2.4	0.5	-0.4	6.0	-50.0	75
SRC/DEST CK1	2.4	0.5	-0.4	6.0	-50.0	75
SRC A BUS EN	2.4	0.5	-0.4	6.0	-50.0	50
DSABLE ALU O	2.4	0.5	-0.4	6.0	-50.0	75
CP CLK (2)	2.4	0.5	-0.4	6.0	-50.0	50
CB ATR	2.4	0.5	-0.4	6.0	-101.0	125
CB REQUEST 1	2.4	0.5	-0.4	6.0	-60.0	50
ANY SUB GNT	2.4	0.5	-0.4	6.0	-60.0	50
SUB GRANT 0	2.4	0.5	-0.4	6.0	-60.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
SUB GRANT 1	2.4	0.5	-0.4	6.0	-60.0	75
SUB GRANT 2	2.4	0.5	-0.4	6.0	-60.0	75
SUB GRANT 3	2.4	0.5	-0.4	6.0	-60.0	100
ANY GNT 1	2.4	0.5	-0.4	6.0	-60.0	75

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	BUS CLK (1) to CB ADDR TMOUT	-	-	-	42	42	42	nS	12
1	tpd	BUS CLK (1) to CB ATR	-	-	-	42	42	42	nS	12
2	tpd	C4 CLK to IR15 A ENBL	-	-	-	27	27	27	nS	12
3	tpd	BUS CLK (1) to ANY MAIN GNT	-	-	-	48	48	48	nS	12
3	tpd	BUS CLK (1) to ANY MAIN GNT	-	-	-	48	48	48	nS	12
4	tpd	BUS CLK (1) to CB GRANT 3	-	-	-	26	26	26	nS	12
4	tpd	BUS CLK (1) to CB GRANT 2	-	-	-	26	26	26	nS	12
4	tpd	BUS CLK (1) to CB GRANT 1	-	-	-	26	26	26	nS	12
4	tpd	BUS CLK (1) to CB GRANT 0	-	-	-	26	26	26	nS	12
5	tpd	1.6 uS CLK to DTA TMOUT FF	-	-	-	54	54	54	nS	12
6	tpd	BUS CLK (1) to DTA TMOUT	-	-	-	41	41	41	nS	12
7	tpd	EXT CLK to CLK 20 MHZ A	-	-	-	24	24	24	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tpd	EXT CLK to CLK 20 MHZ B	-	-	-	24	24	24	nS	12
7	tpd	EXT CLK to CLK 20 MHZ C	-	-	-	24	24	24	nS	12
7	tpd	EXT CLK to CLK 20 MHZ D	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (1)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (2)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (3)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (4)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (5)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (6)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (7)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (8)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (9)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (10)	-	-	-	24	24	24	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tpd	BUS CLK (1) to SUB GRANT 0	-	-	-	26	26	26	nS	12
9	tpd	BUS CLK (1) to SUB GRANT 1	-	-	-	26	26	26	nS	12
9	tpd	BUS CLK (1) to SUB GRANT 2	-	-	-	26	26	26	nS	12
9	tpd	BUS CLK (1) to SUB GRANT 3	-	-	-	26	26	26	nS	12
10	tpd	BUS CLK (1) to ANY SUB GNT	-	-	-	48	48	48	nS	12
10	tpd	BUS CLK (1) to ANY GNT 1	-	-	-	48	48	48	nS	12
11	tpd	BUS CLK (1) to CB REQUEST 1	-	-	-	28	28	28	nS	12
12	tpd	MASTR CB CK to CP CLK (1)	-	-	-	25	25	25	nS	12
12	tpd	MASTR CB CK to CP CLK (2)	-	-	-	25	25	25	nS	12
12	tpd	MASTR CB CK to SRC/DEST CK2	-	-	-	25	25	25	nS	12
12	tpd	MASTR CB CK to SRC/DEST CK1	-	-	-	25	25	25	nS	12
12	tpd	MASTR CB CK to SRC A BUS EN	-	-	-	25	25	25	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	tpd	CP CLK to CP CLK (1)	-	-	-	25	25	25	nS	12
12	tpd	CP CLK to CP CLK (2)	-	-	-	25	25	25	nS	12
12	tpd	CP CLK to SRC/DEST CK2	-	-	-	25	25	25	nS	12
12	tpd	CP CLK to SRC/DEST CK1	-	-	-	25	25	25	nS	12
12	tpd	CP CLK to SRC A BUS EN	-	-	-	25	25	25	nS	12
13	tpd	SRC/DEST CLK to SRC B BUS EN	-	-	-	44	44	44	nS	12
14	tpd	INS REG 32 to DSABLE ALU 0	-	-	-	19	19	19	nS	12
14	tpd	INS REG 33 to DSABLE ALU 0	-	-	-	19	19	19	nS	12
14	tpd	INS REG 30 to DSABLE ALU 0	-	-	-	19	19	19	nS	12
14	tpd	INS REG 29 to DSABLE ALU 0	-	-	-	19	19	19	nS	12
14	tpd	F3 to DSABLE ALU 0	-	-	-	19	19	19	nS	12
14	tpd	A2 to DSABLE ALU 0	-	-	-	19	19	19	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
15	tpd	BUS CLK (1) to 1.6 uS CLK	-	-	-	35	35	35	nS	12
16	tpd	BUS CLK (1) to CLR MN GNT H	-	-	-	49	49	49	nS	12
17	tpd	BUS CLK (1) to CB ADR TMOUT	-	-	-	33	33	33	nS	12
17	tpd	BUS CLK (1) to CB ATR	-	-	-	33	33	33	nS	12
18	tsu	CB SUB REQ 1 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB SUB REQ 2 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB SUB REQ 3 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB REQUEST 0 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB REQUEST 1 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB REQUEST 2 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB REQUEST 3 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB SUB REQ 0 to BUS CLK (1)	13	13	13	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
19	th	BUS CLK (1) to CB ATR	11	11	11	-	-	-	nS	12
20	tsu	CB WR EN to BUS CLK (1)	24	24	24	-	-	-	nS	12
21	th	BUS CLK (1) to CB WR EN	8	8	8	-	-	-	nS	12
22	tsu	CB ORIG RDY to BUS CLK (1)	1	1	1	-	-	-	nS	12
22	tsu	DTA TERM RDY to BUS CLK (1)	1	1	1	-	-	-	nS	12
22	tsu	CB BUSY to BUS CLK (1)	1	1	1	-	-	-	nS	12
23	th	BUS CLK (1) to CB ORIG RDY	12	12	12	-	-	-	nS	12
23	th	BUS CLK (1) to DTA TERM RDY	12	12	12	-	-	-	nS	12
23	th	BUS CLK (1) to CB BUSY	12	12	12	-	-	-	nS	12
24	tsu	DTA TMOUT FF to BUS CLK (1)	12	12	12	-	-	-	nS	12
25	th	BUS CLK (1) to DTA TMOUT FF	10	10	10	-	-	-	nS	12
26	th	BUS CLK (1) to CB GRANT	11	11	11	-	-	-	nS	12
26	th	BUS CLK (1) to SUB ORIG RDY	11	11	11	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
27	tsu	MAE LKUP ACT to BUS CLK (1)	11	11	11	-	-	-	nS	12
28	th	BUS CLK (1) to MAE LKUP ACT	2	2	2	-	-	-	nS	12
29	tsu	FUNCT BIT 2 to BUS CLK (1)	-5	-5	-5	-	-	-	nS	12
30	th	BUS CLK (1) to FUNCT BIT 2	40	40	40	-	-	-	nS	12

6.4 Module function. The bus arbitrator determines what user Central Processor (CP) or Input Output Controllers (IOCs) has control of the common bus. The bus arbitrator module contains a sub arbitrator, main arbitrator, and the system clock distributor. The sub arbitrator determines what IOC uses the sub bus between the IOCs and the Memory Address Expansion (MAE), key code DWF, module. When the sub bus is requested, the sub arbitrator grants the sub bus to the IOC with the highest priority (IOC 0 highest, IOC 3 lowest) and initiates an IOC request to the main arbitrator to use the common bus. If the MRP does not contain an MAE module, the sub arbitrator processes the request in the same manner except the IOC does not use MAE. The main arbitrator grants control of the common bus to the requester with the highest priority. The IOC request from the sub arbitrator has the highest priority and the CP request the lowest. When the request is granted and the bus slave addressed by the bus master (IOC or CP) does not respond within 20 microseconds, an address timeout is generated causing an interrupt. A data timeout is generated when the bus slave does not respond during a data transfer. The data timeout results in an interrupt. The system clock inverts and distributes the clock

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pulses generated on the Arithmetic Logic Unit (ALU) module, key code BWE, to the CP modules to provide timing.

6.4.1 Sub arbitrator circuits. The sub arbitrator circuits consist of the sub request latch, sub request priority, sub grant latch, sub grant detector, hold control, and sub arbitrator control. The sub arbitrator interrogates the IOCs for requests to use MAE, and when requests are received, grants priority to the requester with the highest priority (IOC 0 highest, IOC 3 lowest). The sub arbitrator sends the sub grant signal to the IOC allowing it to use MAE and initiates a request to use the common bus. If the Military Reconfigurable Processor (MRP) does not contain an MAE module, the sub arbitrator processes the request in the same manner except the IOC does not use MAE. The sub request latch holds sub requests from four requesters and is loaded by the bus clock. The sub request priority circuits determine priority for the sub requests held in the latch. The assigned priority is from sub request 0 (highest) to sub request 3 (lowest). The sub request with the highest priority is honored and sets the corresponding sub grant latch bit. The 4-bit sub grant latch holds the sub grant line to the selected IOC low (active) for two bus clock cycles (100 ns). The sub grant detector activates hold control to disable the sub grant signal. Hold control disables priority until the main arbitrator has assigned the IOC control of the common bus and the addressed device has responded with the Address Terminator Ready (ATR) signal or a master clear occurs. Hold control then enables priority to start the next scan. Sub arbitrator control generates an IOC Main Bus Request signal to request control of the common bus. The request is generated when the IOC sends a Sub Arbitrator Originator Ready (OR) signal and the MAE sends a Request signal. The request is terminated when the main arbitrator grants the request.

6.4.2 Main arbitrator circuits. The main arbitrator circuits consist of the main request latch, main request priority, main grant latch, main grant detector, and hold control. The main arbitrator grants control of the common bus to the requester with the highest priority. The main request latch holds requests from four requesters and is loaded by the bus clock. The main request priority circuits determine priority for the main requests held in the latch. The highest assigned priority is main request 0 (IOC main bus request from sub arbitrator) and the lowest is main request 3 (CP). The request with the highest priority is honored and sets the corresponding main grant latch bit. The 4-bit main grant latch holds the main grant line to the selected requester low (active) for two bus

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clock cycles (100 ns) allowing the requester control of the common bus. Note that the main grant 0 line is common to all IOCs and MAE. The main grant detector activates hold control to disable the main priority circuit. This also clears the main grant latch dropping the main grant signal. Hold control disables priority until an ATR is received and the Busy signal is removed, or an address timeout occurs, or a Busy signal was not received within 250 nanoseconds after ATR, or a master clear occurs. Hold control then enables priority to start the next scan.

6.4.3 Address timeout circuits. The circuits consist of an address timer and address timeout sequencer. The circuits generate an address timeout when an ATR is not received within approximately 20 ± 4 microseconds after the main grant signal is active. The address time starts when a main grant is detected. The address timer consists of a one-shot multivibrator and flip-flop. The multivibrator outputs a low pulse for approximately 20 microseconds. If an ATR is received or a master clear occurs within the 20 microseconds, the time is cleared and an address timeout is not generated. If the ATR is not received, the timer activates the address sequencer. The address sequencer places the Address Timeout signal on the common sub and 100 nanoseconds later places an ATR on the common bus. The Address Timeout is on the bus for four clock cycles (200 ns) and the ATR for one clock cycle. The timeout causes the bus master to generate an interrupt.

6.4.4 Data timeout circuits. The circuits consist of a data timer and data timeout sequencer. Data timeouts are caused by one of the following events:

(a) If Busy signal is active for approximately 13 ± 3 microseconds.

(b) When Busy signal goes inactive and a Data Terminator Ready (DTR) does not occur within one clock cycle.

(c) If ATR has occurred and Busy signal has been inactive for 250 ns.

The data timer times the Busy signal. The timer consists of a one-shot multivibrator and flip-flop. The multivibrator outputs a low pulse for approximately 13 microseconds after being triggered by the Busy signal. If the Busy signal goes inactive and a DTR is received, or a data timeout occurs (from another event), or a master clear occurs within the 13 microseconds,

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the timer is cleared. If the timer is not cleared, a data timeout is generated. The data sequencer is activated when any one of the three events occur. The sequencer places the Data Timeout signal on the common bus for five clock cycles (250 ns). The timeout causes the bus master to generate an interrupt and the bus slave to clear itself.

6.4.5 Function decoder and control. The function decoder and control circuits decode the common bus function code and interpret the common bus control signals to provide control for the bus arbitrator module.

6.4.6 System clock. The system clock inverts and distributes the clock pulses generated on the ALU module to the CP modules to provide timing.

6.5 Description of I/O pins.

Signal name	Pin number	I/O	Description
CB SUB REQ 1	2	I	CB SUB REQUEST 1 (IOC2)
CB SUB REQ 0	3	I	CB SUB REQUEST 0 (IOC3)
CB REQUEST 3	4	I	CB REQUEST 3 (CP)
CB REQUEST 2	5	I	CB REQUEST 2 (SPARE)
CB REQUEST 1	6	I	CB REQUEST 1 (IOC)
CB REQUEST 0	7	I	CB REQUEST 0 (DMA)
INS REG 32	14	I	MICRO INSTR RGTR 32 (F0)
INS REG 33	15	I	MICRO INSTR RGTR 33 (F1)
INS REG 30	16	I	MICRO INSTR RGTR 30 (C1)
INS REG 29	17	I	MICRO INSTR RGTR 29 (C0)
DTA TMOUT FF	18	I	DATA TIMEOUT FF
EN EXT CLK	19	I	EN EXT CLK

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6.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
EXT CLK	20	I	EXT CLK
MICRO IR 15	29	I	MICRO IR 15
EN IC TMOUT	30	I	EN IC TMOUT
1.6 uS CLK	42	I	1.6 MICROSEC CLK
CB SUB REQ 2	44	I	CB SUB REQ 2 (IOC1)
CB SUB REQ 3	45	I	CB SUB REQ 3 (IOC0)
CP CLK	54	I	CP CLK
MAINT PRCSR	58	I	MAINT PRCSR F SCE B BUS ENBL
F3	59	I	F3
MASTR CB CK	61	I	MASTER CB CLK (CPALU)
A2	65	I	A2
C4 CLK	66	I	C4 CLK
SRC/DEST CLK	67	I	SOURCE/DEST CLK
FUNCT BIT 2	74	I	FUNCTION BIT 2
BUS CLK (1)	76	I	BUS CLK (1)
CB ORIG RDY	85	I	CB ORIGINATOR RDY
DTA TERM RDY	86	I	CB DATA TERMINATOR RDY
CB MA CLR	87	I	CB MA CLR
CB BUSY	88	I	CB BUSY
CB GRANT	91	I	CB GRANT 1 (IOC)

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6.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MAE LKUP ACT	92	I	MAE LOOKUP ACTIVE
SUB ORG RDY	93	I	SUBARBITRATOR ORIGINATOR RDY
CB WR EN	95	I	CB WRITE ENABLE
CB GRANT 3	8	O	CB GRANT 3 (CP)
CB GRANT 2	9	O	CB GRANT 2 (SPARE)
CB GRANT 1	11	O	CB GRANT 1 (IOC)
CB GRANT 0	12	O	CB GRANT 0 (DMA)
ANY MAIN GNT	13	O	ANY MAIN GRANT
19HZ ICT B	21	O	19 HZ ICTO CLK (B)
CLK 20 MHZ B	22	O	CLK 20 MHZ (B)
CLK 20 MHZ D	23	O	CLK 20 MHZ (D)
19HZ DSTK CK	24	O	19 HZ DEADSTICK CLK
CLK 20 MHZ C	26	O	CLK 20 MHZ (C)
19HZ ICT A	27	O	19 HZ ICTO CLK (A)
ANY MAIN GNT	28	O	ANY MAIN GRANT
IR15 A ENBL	32	O	MICRO IR 15 and SCE A ENBL
CB CLK (4)	33	O	CB CLK (4)
CB CLK (5)	34	O	CB CLK (5)
CB CLK (6)	35	O	CB CLK (6)
CB CLK (7)	36	O	CB CLK (7)

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6.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
25.6 uS CLK	43	O	25.6 MICROSEC CLK
CLK 20 MHZ A	46	O	CLK 20 MHZ (A)
CLR MN GNT H	47	O	CLR MN GRANT HOLD
ACTIV PULLUP	48	O	ACTIVE PULLUP
CB CLK (8)	37	O	CB CLK (8)
CB CLK (9)	38	O	CB CLK (9)
CB CLK (10)	39	O	CB CLK (10)
DTA TMOUT FF	41	O	DATA TIMEOUT FF
CB DTA TMOUT	49	O	CB DATA TIMEOUT
CB ADR TMOUT	50	O	CB ADDR TIMEOUT
CB CLK (3)	53	O	CB CLK (3)
1.6 uS CLK	56	O	1.6 MICROSEC CLK
SRC B BUS EN	63	O	SOURCE B BUS ENBL
CP CLK (1)	69	O	CP CLK (1)
CB CLK (2)	72	O	CB CLK (2)
CB CLK (1)	73	O	CB CLK (1)
SRC/DEST CK2	77	O	SOURCE/DEST CLK 2
SRC/DEST CK1	79	O	SOURCE/DEST CLK 1
SRC A BUS EN	80	O	SOURCE A BUS ENBL
DSABLE ALU O	82	O	DISABLE ALU OUTPUT

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6.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CP CLK (2)	83	O	CP CLK (2)
CB ATR	84	I/O	CB ATR
CB REQUEST 1	89	O	CB REQUEST 1 (IOC)
ANY SUB GNT	94	O	ANY SUB GRANT
SUB GRANT 0	96	O	SUB GRANT 0
SUB GRANT 1	97	O	SUB GRANT 1
SUB GRANT 2	98	O	SUB GRANT 2
SUB GRANT 3	99	O	SUB GRANT 3
ANY GNT 1	100	O	ANY SUB GRANT

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N713)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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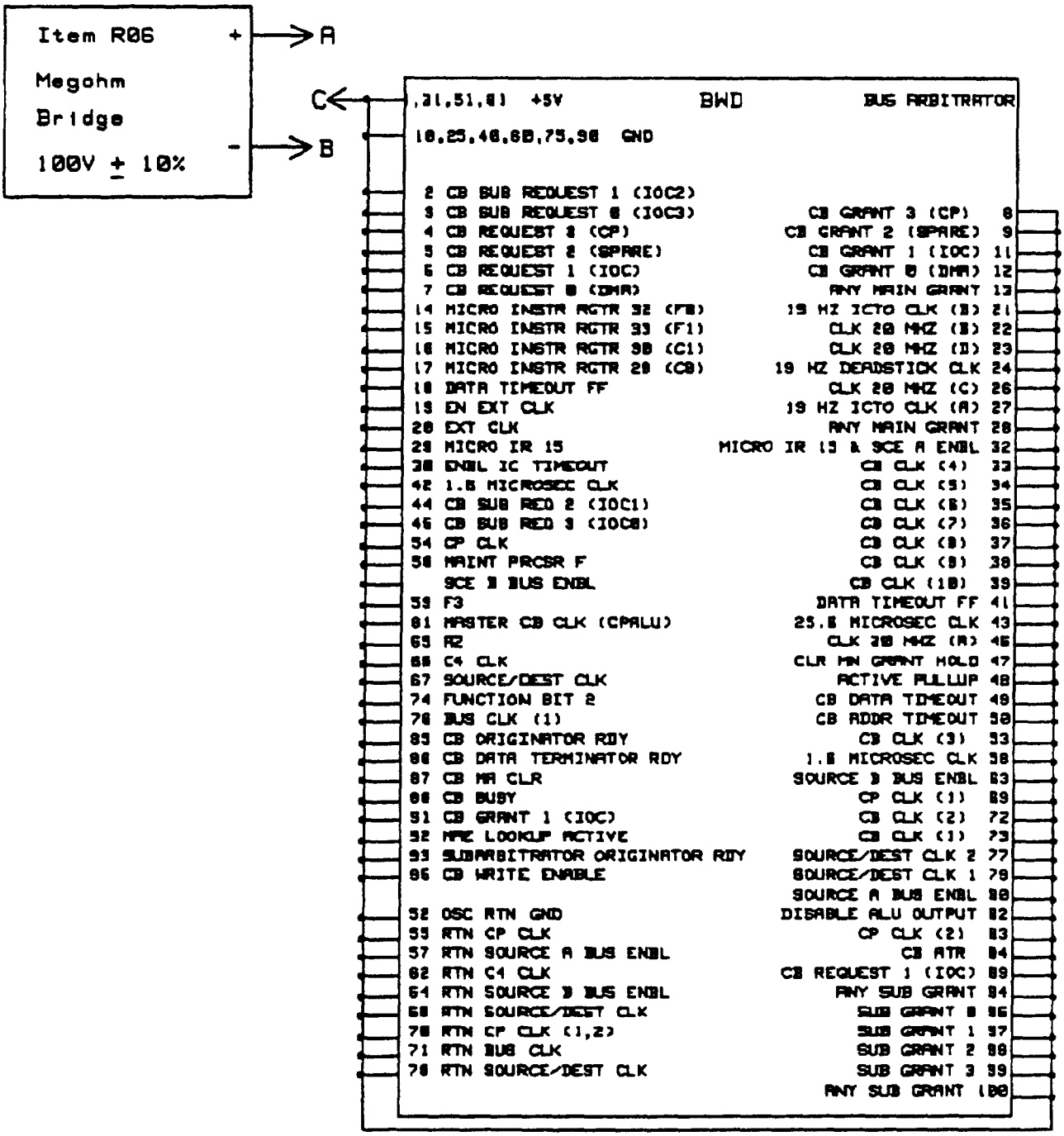


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Signal name	Connect		25 ± 5°C Initial limits		Units	Notes
		A to	B to	Min	Max		
B1	GND	10	25	0	1	Ohm	1-5
B2	GND	10	40				
B3	GND	60	75				
B4	GND	60	90				
B5	+5V	1	31				
B6	+5V	51	81				
B7	OSC RTN GND	60	52				
B8	RTN CP CLK	60	55				
B9	RTN SRC A EN	60	57				
B10	RTN C4 CLK	60	62				
B11	RTN SRC B EN	60	64				
B12	RTN SRC/DEST	60	68				
B13	RTN CK (1,2)	60	70				
B14	RTN BUS CLK	60	71				
B15	RTN SRC/DEST	60	78	0	1	Ohm	1-5

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NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B15

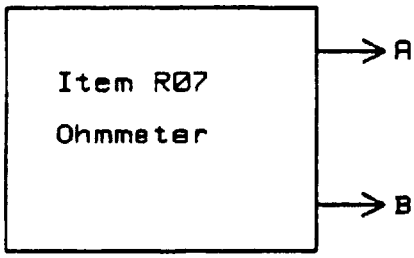
2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,51,81 +5V	BWD	BUS ARBITRATOR	
10,25,40,50,75,90 GND			
2 CB SUB REQUEST 1 (IOC2)			
3 CB SUB REQUEST 0 (IOC3)		CB GRANT 3 (CP)	8
4 CB REQUEST 3 (CP)		CB GRANT 2 (SPARE)	9
5 CB REQUEST 2 (SPARE)		CB GRANT 1 (IOC)	11
6 CB REQUEST 1 (IOC)		CB GRANT 0 (DMA)	12
7 CB REQUEST 0 (DMA)		ANY MAIN GRANT	13
14 MICRO INSTR RGTR 32 (F0)		10 MZ ICTO CLK (B)	21
15 MICRO INSTR RGTR 33 (F1)		CLK 20 MHZ (B)	22
16 MICRO INSTR RGTR 30 (C1)		CLK 20 MHZ (D)	23
17 MICRO INSTR RGTR 29 (C0)		19 MZ DEADSTICK CLK	24
18 DATA TIMEOUT FF		CLK 20 MHZ (C)	26
19 EN EXT CLK		10 MZ ICTO CLK (A)	27
20 EXT CLK		ANY MAIN GRANT	28
29 MICRO IR 15		MICRO IR 15 & SCE A ENBL	32
30 ENBL IC TIMEOUT		CB CLK (4)	33
42 1.6 MICROSEC CLK		CB CLK (5)	34
44 CB SUB REQ 2 (IOC1)		CB CLK (6)	35
45 CB SUB REQ 3 (IOC0)		CB CLK (7)	36
54 CP CLK		CB CLK (8)	37
58 MAINT PROCR F		CB CLK (9)	38
SCE B BUS ENBL		CB CLK (10)	39
59 F3		DATA TIMEOUT FF	41
61 MASTER CB CLK (CPALU)		25.6 MICROSEC CLK	43
65 A2		CLK 20 MHZ (A)	46
66 C4 CLK		CLR MN GRANT HOLD	47
67 SOURCE/DEST CLK		ACTIVE PULLUP	48
74 FUNCTION BIT 2		CB DATA TIMEOUT	49
76 BUS CLK (1)		CB ADDR TIMEOUT	50
85 CB ORIGINATOR RDY		CB CLK (3)	53
86 CB DATA TERMINATOR RDY		1.6 MICROSEC CLK	58
87 CB MA CLR		SOURCE B BUS ENBL	63
88 CB BUSY		CP CLK (1)	69
91 CB GRANT 1 (IOC)		CB CLK (2)	72
92 MRE LOOKUP ACTIVE		CB CLK (1)	73
93 SUBARBITRATOR ORIGINATOR RDY		SOURCE/DEST CLK 2	77
95 CB WRITE ENABLE		SOURCE/DEST CLK 1	79
		SOURCE A BUS ENBL	80
52 OSC RTN GND		DISABLE ALU OUTPUT	82
55 RTN CP CLK		CP CLK (2)	83
57 RTN SOURCE A BUS ENBL		CB RTR	84
62 RTN C4 CLK		CB REQUEST 1 (IOC)	89
64 RTN SOURCE B BUS ENBL		ANY SUB GRANT	94
68 RTN SOURCE/DEST CLK		SUB GRANT 0	96
70 RTN CP CLK (1,2)		SUB GRANT 1	97
71 RTN BUS CLK		SUB GRANT 2	98
78 RTN SOURCE/DEST CLK		SUB GRANT 3	99
		ANY SUB GRANT	100

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Notes
			Positive lead to	Negative lead to		
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	3.6 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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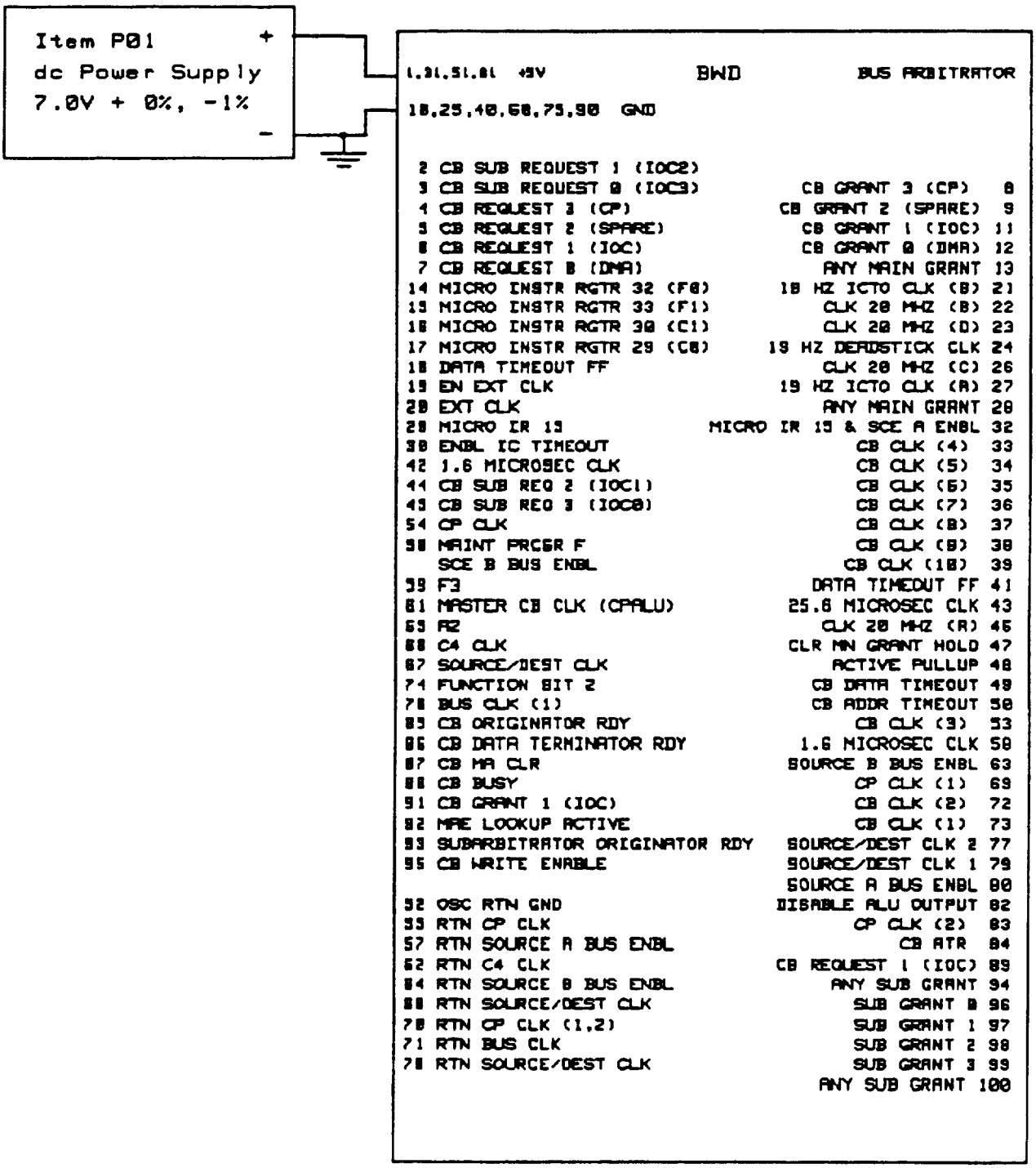


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	CB SUB REQ 1	2	5.5	-0.027	0.122	mA	1-3
D2	CB SUB REQ 0	3					
D3	CB REQUEST 3	4					
D4	CB REQUEST 2	5					
D5	CB REQUEST 1	6					
D6	CB REQUEST 0	7	5.5	-0.027	0.122	mA	1-3
D7	INS REG 32	14	5.5	0.0	0.1	mA	1-3
D8	INS REG 33	15					
D9	INS REG 30	16					
D10	INS REG 29	17					
D11	DTA TMOUT FF	18	5.5	0.0	0.1	mA	1-3
D12	EN EXT CLK	19	5.5	-0.05	2.04	mA	1-3
D13	EXT CLK	20	5.5	0.0	1.0	mA	1-3
D14	MICRO IR 15	29	5.5	0.0	0.1	mA	1-3
D15	EN IC TMOUT	30	5.5	-0.027	0.122	mA	1-3
D16	1.6 uS CLK	42	5.5	0.0	0.2	mA	1-3
D17	CB SUB REQ 2	44	5.5	-0.027	0.122	mA	1-3
D18	CB SUB REQ 3	45	5.5	-0.027	0.122	mA	1-3
D19	CP CLK	54	5.5	0.0	1.0	mA	1-3
D20	MAINT PRCSR	58	5.5	-0.027	0.122	mA	1-3
D21	F3	59	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D22	MASTR CB CK	61	5.5	0.0	2.0	mA	1-3
D23	A2	65	5.5	0.0	0.1	mA	1-3
D24	C4 CLK	66	5.5	0.0	2.0	mA	1-3
D25	SRC/DEST CLK	67	5.5	0.0	1.0	mA	1-3
D26	FUNCT BIT 2	74	5.5	0.0	0.1	mA	1-3
D27	BUS CLK (1)	76	5.5	0.0	2.0	mA	1-3
D28	CB ORIG RDY	85	5.5	-0.027	0.122	mA	1-3
D29	DTA TERM RDY	86	5.5	-0.027	0.122	mA	1-3
D30	CB MA CLR	87	5.5	0.0	1.0	mA	1-3
D31	CB BUSY	88	5.5	-0.027	0.122	mA	1-3
D32	CB GRANT	91	5.5	0.0	0.1	mA	1-3
D33	MAE LKUP ACT	92	5.5	-0.027	0.122	mA	1-3
D34	SUB ORG RDY	93					
D35	CB WR EN	95	5.5	-0.027	0.122	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D35
Power supply current limit	3.0 A	NA	D1-D35
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D35
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D35 D1-D35
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 7.45 uA ± 5.25 uA ± 105.0 uA ± 50.3 uA ± 5.25 uA ± 7.45 uA ± 10.3 uA ± 7.45 uA ± 50.3 uA ± 7.45 uA ± 5.25 uA ± 100.0 uA ± 5.25 uA ± 100.0 uA ± 50.3 uA ± 5.25 uA ± 100.0 uA ± 7.45 uA ± 50.3 uA ± 7.45 uA ± 5.25 uA ± 7.45 uA	D1-D6 D7-D11 D12 D13 D14 D15 D16 D17,D18 D19 D20 D21 D22 D23 D24 D25 D26 D27 D28,D29 D30 D31 D32 D33-D35

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	CB GRANT 3	8	0.0	-150.0	-60.0	mA	1-4
E2	CB GRANT 2	9					
E3	CB GRANT 1	11					
E4	CB GRANT 0	12					
E5	ANY MAIN GNT	13	0.0	-150.0	-60.0	mA	1-4
E6	19HZ ICT B	21	0.0	-225.0	-50.0	mA	1-4
E7	CLK 20 MHZ B	22					
E8	CLK 20 MHZ D	23					
E9	19HZ DSTK CK	24					
E10	CLK 20 MHZ C	26					
E11	19HZ ICT A	27	0.0	-225.0	-50.0	mA	1-4
E12	ANY MAIN GNT	28	0.0	-100.0	-40.0	mA	1-4
E13	IR15 A ENBL	32	0.0	-150.0	-60.0	mA	1-4
E14	CB CLK (4)	33	0.0	-225.0	-50.0	mA	1-4
E15	CB CLK (5)	34					
E16	CB CLK (6)	35					
E17	CB CLK (7)	36					
E18	CB CLK (8)	37					
E19	CB CLK (9)	38					
E20	CB CLK (10)	39	0.0	-225.0	-50.0	mA	1-4
E21	DTA TMOUT FF	41	0.0	-150.0	-60.0	mA	1-4

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Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	25.6 uS CLK	43	0.0	-120.0	-20.0	mA	1-4
E23	CLK 20 MHZ A	46	0.0	-225.0	-50.0	mA	1-4
E24	CLR MN GNT H	47	0.0	-150.0	-60.0	mA	1-4
E25	ACTIV PULLUP	48	0.0	-325.0	-100.0	mA	1-4
E26	CB DTA TMOUT	49	0.0	-325.0	-100.0	mA	1-4
E27	CB ADR TMOUT	50	0.0	-326.0	-101.0	mA	1-4
E28	CB CLK (3)	53	0.0	-225.0	-50.0	mA	1-4
E29	1.6 uS CLK	56	0.0	-100.0	-40.0	mA	1-4
E30	SRC B BUS EN	63	0.0	-100.0	-40.0	mA	1-4
E31	CP CLK (1)	69	0.0	-225.0	-50.0	mA	1-4
E32	CB CLK (2)	72					
E33	CB CLK (1)	73					
E34	SRC/DEST CK2	77					
E35	SRC/DEST CK1	79					
E36	SRC A BUS EN	80	0.0	-225.0	-50.0	mA	1-4
E37	DSABLE ALU O	82	0.0	-150.0	-60.0	mA	1-4
E38	CP CLK (2)	83	0.0	-225.0	-50.0	mA	1-4
E39	CB ATR	84	0.0	-326.0	-101.0	mA	1-4
E40	CB REQUEST 1	89	0.0	-150.0	-60.0	mA	1-4
E41	ANY SUB GNT	94					
E42	SUB GRANT 0	96	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E43	SUB GRANT 1	97	0.0	-150.0	-60.0	mA	1-4
E44	SUB GRANT 2	98					
E45	SUB GRANT 3	99					
E46	ANY GNT 1	100	0.0	-150.0	-60.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E46
Power supply current limit	3.0 A	NA	E1-E46
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E46
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E46
low, (VL)	0.0 V	± 100 mV	E1-E46

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:25Z
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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 4.50 mA	E1-E5
	NA	± 8.75 mA	E6-E11
	NA	± 3.0 mA	E12
	NA	± 4.50 mA	E13
	NA	± 8.75 mA	E14-E20
	NA	± 4.50 mA	E21
	NA	± 5.0 mA	E22
	NA	± 8.75 mA	E23
	NA	± 4.50 mA	E24
	NA	± 11.3 mA	E25-E27
	NA	± 8.75 mA	E28
	NA	± 3.0 mA	E29,E30
	NA	± 8.75 mA	E31-E36
	NA	± 4.5 mA	E37
	NA	± 8.75 mA	E38
	NA	± 11.3 mA	E39
	NA	± 4.50 mA	E40-E46

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

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NOTES: - Continued.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	CB SUB REQ 1	2	2.4	-1.66	-0.544	mA	1-3
F2	CB SUB REQ 0	3					
F3	CB REQUEST 3	4					
F4	CB REQUEST 2	5					
F5	CB REQUEST 1	6					
F6	CB REQUEST 0	7	2.4	-1.66	-0.544	mA	1-3
F7	INS REG 32	14	2.4	-0.6	0.02	mA	1-3
F8	INS REG 33	15					
F9	INS REG 30	16					
F10	INS REG 29	17					
F11	DTA TMOUT FF	18	2.4	-0.6	0.02	mA	1-3
F12	EXT CLK	20	2.4	-2.0	0.05	mA	1-3
F13	MICRO IR 15	29	2.4	-0.6	0.02	mA	1-3
F14	EN IC TMOUT	30	2.4	-1.66	-0.544	mA	1-3
F15	1.6 uS CLK	42	2.4	-0.8	0.04	mA	1-3
F16	CB SUB REQ 2	44	2.4	-1.66	-0.544	mA	1-3
F17	CB SUB REQ 3	45	2.4	-1.66	-0.544	mA	1-3
F18	CP CLK	54	2.4	-2.0	0.05	mA	1-3
F19	MAINT PRCSR	58	2.4	-1.66	-0.544	mA	1-3
F20	F3	59	2.4	-0.6	0.02	mA	1-3
F21	MASTR CB CK	61	2.4	-4.0	0.1	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	A2	65	2.4	-0.6	0.02	mA	1-3
F23	C4 CLK	66	2.4	-4.0	0.1	mA	1-3
F24	SRC/DEST CLK	67	2.4	-2.0	0.05	mA	1-3
F25	FUNCT BIT 2	74	2.4	-0.6	0.02	mA	1-3
F26	BUS CLK (1)	76	2.4	-4.0	0.1	mA	1-3
F27	CB ATR	84	2.4	-2.91	-0.454	mA	1-3
F28	CB ORIG RDY	85	2.4	-1.66	-0.544	mA	1-3
F29	DTA TERM RDY	86	2.4	-1.66	-0.544	mA	1-3
F30	CB MA CLR	87	2.4	-2.0	0.05	mA	1-3
F31	CB BUSY	88	2.4	-1.66	-0.544	mA	1-3
F32	CB GRANT	91	2.4	-0.6	0.02	mA	1-3
F33	MAE LKUP ACT	92	2.4	-1.66	-0.544	mA	1-3
F34	SUB ORG RDY	93	″	″	″	″	″
F35	CB WR EN	95	2.4	-1.66	-0.544	mA	1-3
F36	EN EXT CLK	19	2.4	-4.52	-1.02	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:25Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F36
Power supply current limit	3.0 A	NA	F1-F36
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F36
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F36
low, (VL)	0.0 V	± 100 mV	F1-F36
Input current measurement	NA	± 55.8 uA	F1-F6
	NA	± 31.0 uA	F7-F11
	NA	± 103.0 uA	F12
	NA	± 31.0 uA	F13
	NA	± 55.8 uA	F14
	NA	± 42.0 uA	F15
	NA	± 55.8 uA	F16,F17
	NA	± 103.0 uA	F18
	NA	± 55.8 uA	F19
	NA	± 31.0 uA	F20
	NA	± 205.0 uA	F21
	NA	± 31.0 uA	F22
	NA	± 205.0 uA	F23
	NA	± 103.0 uA	F24
	NA	± 31.0 uA	F25
	NA	± 205.0 uA	F26
	NA	± 123.0 uA	F27
	NA	± 55.8 uA	F28,F29
	NA	± 103.0 uA	F30
	NA	± 55.8 uA	F31
	NA	± 31.0 uA	F32
	NA	± 55.8 uA	F33-F35
	NA	± 175.0 uA	F36

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	CB SUB REQ 1	2	0.4	-1.66	-0.544	mA	1-3
G2	CB SUB REQ 0	3					
G3	CB REQUEST 3	4					
G4	CB REQUEST 2	5					
G5	CB REQUEST 1	6					
G6	CB REQUEST 0	7	0.4	-1.66	-0.544	mA	1-3
G7	INS REG 32	14	0.4	-0.6	0.02	mA	1-3
G8	INS REG 33	15					
G9	INS REG 30	16					
G10	INS REG 29	17					
G11	DTA TMOUT FF	18	0.4	-0.6	0.02	mA	1-3
G12	EN EXT CLK	19	0.4	-4.52	-1.02	mA	1-3
G13	EXT CLK	20	0.4	-2.0	0.05	mA	1-3
G14	MICRO IR 15	29	0.4	-0.6	0.02	mA	1-3
G15	EN IC TMOUT	30	0.4	-1.66	-0.544	mA	1-3
G16	1.6 uS CLK	42	0.4	-0.8	0.04	mA	1-3
G17	CB SUB REQ 2	44	0.4	-1.66	-0.544	mA	1-3
G18	CB SUB REQ 3	45	0.4	-1.66	-0.544	mA	1-3
G19	CP CLK	54	0.4	-2.0	0.05	mA	1-3
G20	MAINT PRCSR	58	0.4	-1.66	-0.544	mA	1-3
G21	F3	59	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G22	MASTR CB CK	61	0.4	-4.0	0.1	mA	1-3
G23	A2	65	0.4	-0.6	0.02	mA	1-3
G24	C4 CLK	66	0.4	-4.0	0.1	mA	1-3
G25	SRC/DEST CLK	67	0.4	-2.0	0.05	mA	1-3
G26	FUNCT BIT 2	74	0.4	-0.6	0.02	mA	1-3
G27	BUS CLK (1)	76	0.4	-4.0	0.1	mA	1-3
G28	CB ATR	84	0.4	-2.91	-0.454	mA	1-3
G29	CB ORIG RDY	85	0.4	-1.66	-0.544	mA	1-3
G30	DTA TERM RDY	86	0.4	-1.66	-0.544	mA	1-3
G31	CB MA CLR	87	0.4	-2.0	0.05	mA	1-3
G32	CB BUSY	88	0.4	-1.66	-0.544	mA	1-3
G33	CB GRANT	91	0.4	-0.6	0.02	mA	1-3
G34	MAE LKUP ACT	92	0.4	-1.66	-0.544	mA	1-3
G35	SUB ORG RDY	93					
G36	CB WR EN	95	0.4	-1.66	-0.544	mA	1-3

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Check the source to verify that this is the current version before use.

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G36
Power supply current limit	3.0 A	NA	G1-G36
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G36
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G36
low, (VL)	0.0 V	± 100 mV	G1-G36
Input current measurement	NA	± 55.8 uA	G1-G6
	NA	± 31.0 uA	G7-G11
	NA	± 175.0 uA	G12
	NA	± 103.0 uA	G13
	NA	± 31.0 uA	G14
	NA	± 55.8 uA	G15
	NA	± 42.0 uA	G16
	NA	± 55.8 uA	G17,G18
	NA	± 103.0 uA	G19
	NA	± 55.8 uA	G20
	NA	± 41.0 uA	G21
	NA	± 205.0 uA	G22
	NA	± 31.0 uA	G23
	NA	± 205.0 uA	G24
	NA	± 103.0 uA	G25
	NA	± 31.0 uA	G26
	NA	± 205.0 uA	G27
	NA	± 123.0 uA	G28
	NA	± 55.8 uA	G29,G30
	NA	± 103.0 uA	G31
	NA	± 55.8 uA	G32
	NA	± 31.0 uA	G33
	NA	± 55.8 uA	G34-G36

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	CB GRANT 3	8	-0.4	2.4	4.5	V	1-4
H2	CB GRANT 2	9					
H3	CB GRANT 1	11					
H4	CB GRANT 0	12					
H5	ANY MAIN GNT	13					
H6	19HZ ICT B	21					
H7	CLK 20 MHZ B	22					
H8	CLK 20 MHZ D	23					
H9	19HZ DSTK CK	24					
H10	CLK 20 MHZ C	26					
H11	19HZ ICT A	27					
H12	ANY MAIN GNT	28					
H13	IR15 A ENBL	32					
H14	CB CLK (4)	33					
H15	CB CLK (5)	34					
H16	CB CLK (6)	35					
H17	CB CLK (7)	36					
H18	CB CLK (8)	37					
H19	CB CLK (9)	38					
H20	CB CLK (10)	39					
H21	DTA TMOUT FF	41	-0.4	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:25Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H22	25.6 uS CLK	43	-0.16	2.4	4.5	V	1-4
H23	CLK 20 MHZ A	46	-0.4	2.4	4.5	V	1-4
H24	CLR MN GNT H	47					
H25	ACTIV PULLUP	48					
H26	CB DTA TMOUT	49					
H27	CB ADR TMOUT	50					
H28	CB CLK (3)	53					
H29	1.6 uS CLK	56					
H30	SRC B BUS EN	63					
H31	CP CLK (1)	69					
H32	CB CLK (2)	72					
H33	CB CLK (1)	73					
H34	SRC/DEST CK2	77					
H35	SRC/DEST CK1	79					
H36	SRC A BUS EN	80					
H37	DSABLE ALU O	82					
H38	CP CLK (2)	83					
H39	CB ATR	84					
H40	CB REQUEST 1	89	-0.4	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	ANY SUB GNT	94	-0.4	2.4	4.5	V	1-4
H42	SUB GRANT 0	96					
H43	SUB GRANT 1	97					
H44	SUB GRANT 2	98					
H45	SUB GRANT 3	99					
H46	ANY GNT 1	100	-0.4	2.4	4.5	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H46
Power supply current limit	3.0 A	NA	H1-H46
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA	± 40 uA ± 16 uA ± 40 uA	H1-H21 H22 H23-H46
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H46 H1-H46
Output voltage measurement	NA	± 105 mV	H1-H46

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

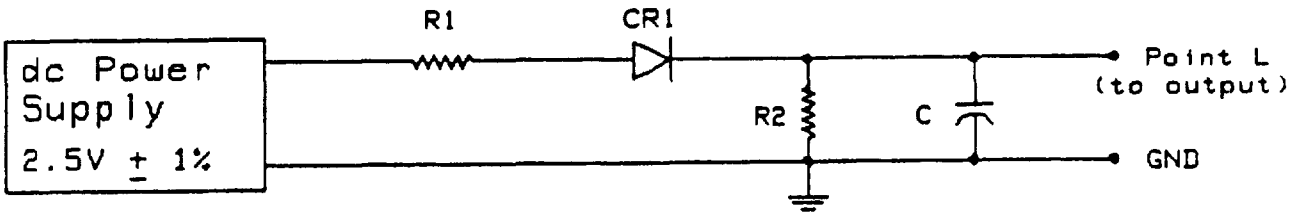
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
8,9,11,12,13,21,22, 23,24,26,27,28,32, 33,34,35,36,37,38, 39,41,46,47,49,50, 53,56,63,69,72,73, 77,79,80,82,83,84, 89,94,96,97,98,99, 100	6200	6.0	0.5
43	16000	1.6	0.5
48	6200	NA	NA

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	CB GRANT 3	8	6.0	0.0	0.5	V	1-4
J2	CB GRANT 2	9					
J3	CB GRANT 1	11					
J4	CB GRANT 0	12					
J5	ANY MAIN GNT	13					
J6	CLK 20 MHZ B	22					
J7	CLK 20 MHZ D	23					
J8	CLK 20 MHZ C	26					
J9	ANY MAIN GNT	28					
J10	IR15 A ENBL	32					
J11	CB CLK (4)	33					
J12	CB CLK (5)	34					
J13	CB CLK (6)	35					
J14	CB CLK (7)	36					
J15	CB CLK (8)	37					
J16	CB CLK (9)	38					
J17	CB CLK (10)	39					
J18	DTA TMOUT FF	41	6.0	0.0	0.5	V	1-4
J19	25.6 uS CLK	43	1.6	0.0	0.5	V	1-4
J20	CLK 20 MHZ A	46	6.0	0.0	0.5	V	1-4
J21	CLR MN GNT H	47	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J22	CB DTA TMOUT	49	6.0	0.0	0.5	V	1-4
J23	CB ADR TMOUT	50					
J24	CB CLK (3)	53					
J25	1.6 uS CLK	56					
J26	SRC B BUS EN	63					
J27	CP CLK (1)	69					
J28	CB CLK (2)	72					
J29	CB CLK (1)	73					
J30	SRC/DEST CK2	77					
J31	SRC/DEST CK1	79					
J32	SRC A BUS EN	80					
J33	DSABLE ALU O	82					
J34	CP CLK (2)	83					
J35	CB ATR	84					
J36	CB REQUEST 1	89					
J37	ANY SUB GNT	94					
J38	SUB GRANT 0	96					
J39	SUB GRANT 1	97					
J40	SUB GRANT 2	98					
J41	SUB GRANT 3	99					
J42	ANY GNT 1	100	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J43	19HZ ICT B	21	6.0	0.0	0.5	V	1-4
J44	19HZ DSTK CK	24					
J45	19HZ ICT A	27	6.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J45
Power supply current limit	3.0 A	NA	J1-J45
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA	± 600 uA ± 160 uA ± 600 uA	J1-J18 J19 J20-J45
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J45 J1-J45
Output voltage measurement	NA	± 25 mV	J1-J45

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	INS REG 32	14	-18.0	-1.2	0.0	V	1-3
L2	INS REG 33	15					
L3	INS REG 30	16					
L4	INS REG 29	17					
L5	DTA TMOUT FF	18					
L6	EXT CLK	20					
L7	MICRO IR 15	29	-18.0	-1.2	0.0	V	1-3
L8	1.6 uS CLK	42	-18.0	-1.5	0.0	V	1-3
L9	CP CLK	54	-18.0	-1.2	0.0	V	1-3
L10	F3	59					
L11	MASTR CB CK	61					
L12	A2	65					
L13	C4 CLK	66					
L14	SRC/DEST CLK	67					
L15	FUNCT BIT 2	74					
L16	BUS CLK (1)	76					
L17	CB MA CLR	87					
L18	CB GRANT	91	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L18
Power supply current limit	NA	NA	L1-L18
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L18
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L18 L1-L18
Input voltage measurement	NA NA NA	± 60 mV ± 75 mV ± 60 mV	L1-L7 L8 L9-L18

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Power supply filter capacitance test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	4.2	8.0	uF	1-4
M2	PSFC	51	5.0	9.4	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 190 nF ± 220 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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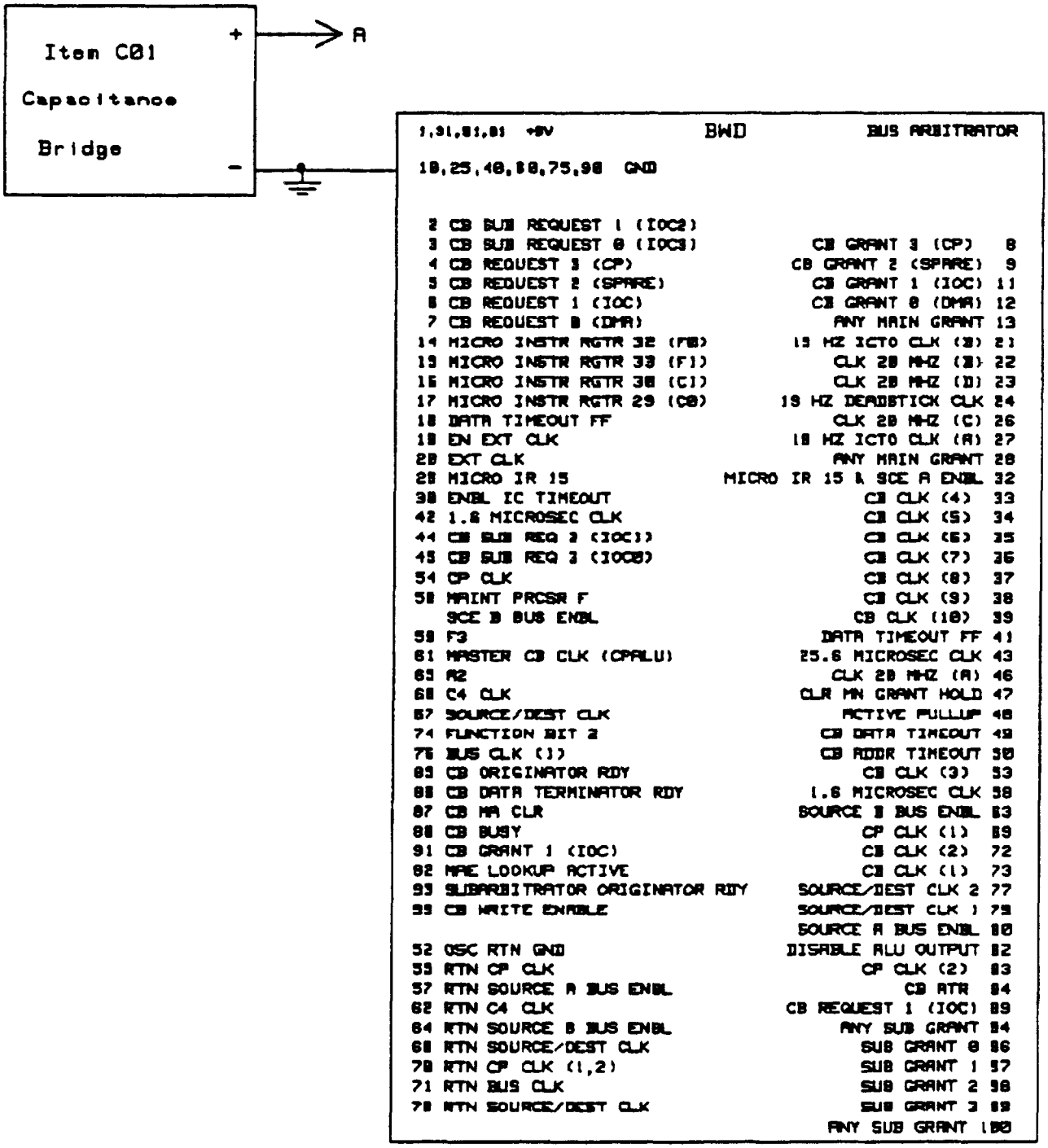


FIGURE 10. Power supply filter capacitance test circuit for table XII.

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TABLE XIII. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
N	Icc	1,31, 51,81	830	2000	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	3.0 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 60 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Functional test.

Test cond	Test description	Notes	Figs.
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	3.0 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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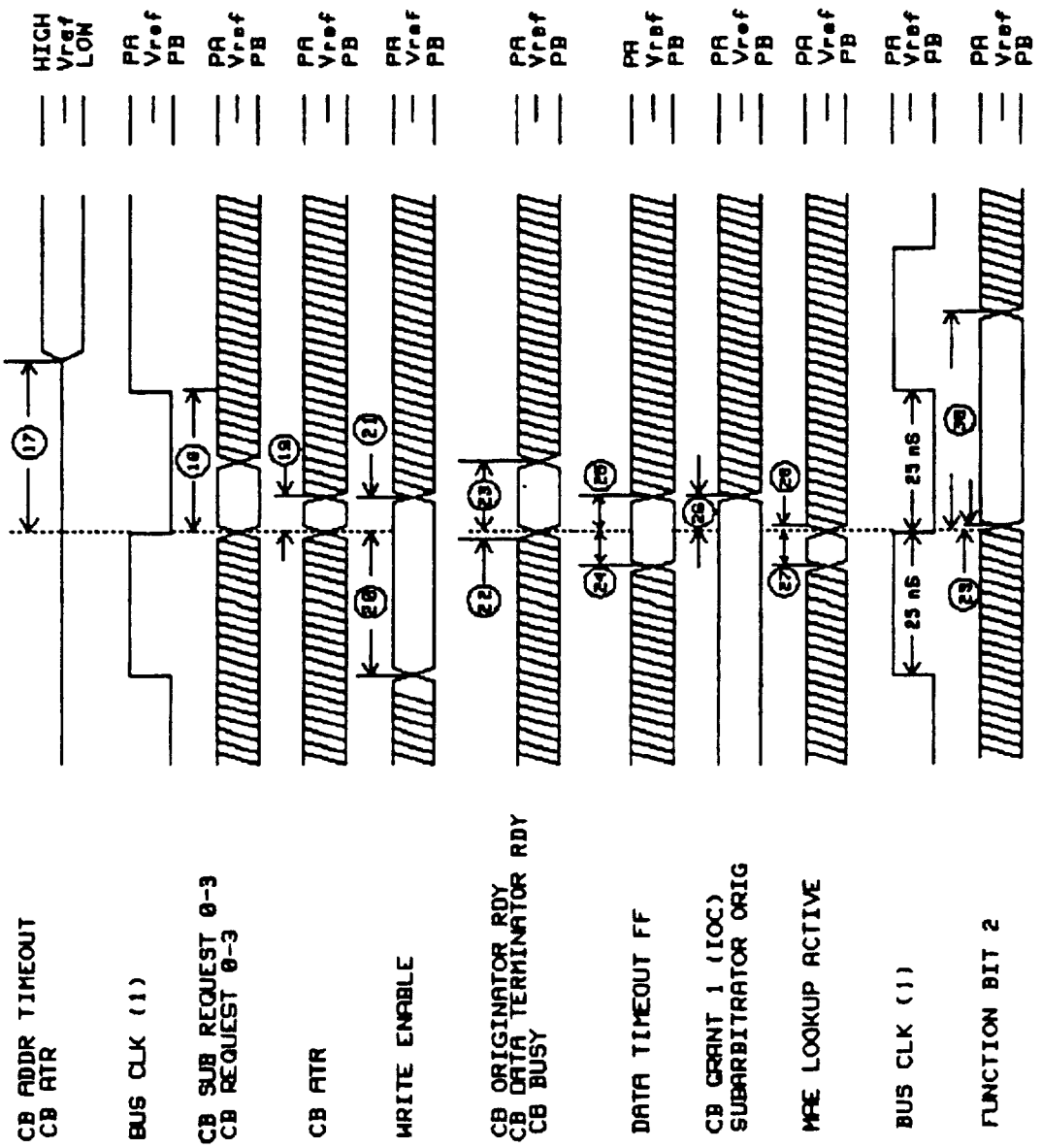


FIGURE 11. Input timing parameters for table XIV.

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NOTES:

1. (18) CB SUB REQ 1, CB REQUEST 3, CB REQUEST 2, CB REQUEST 1, CB REQUEST 0, CB SUB REQUEST 0, CB SUB REQUEST 2, CB SUB REQUEST 3, to BUS CLK (1). Setup time-minimum 13 nS.

2. (19) BUS CLK (1) to CB ATR. Hold time-minimum 11 nS.

3. (20) CB WR EN to BUS CLK (1). Setup time-minimum 24 nS.

4. (21) BUS CLK (1) to CB WR EN. Hold time-minimum 8 nS.

5. (22) CB ORIG RDY to DTA TERM RDY, CB BUSY, BUS CLK (1). Setup time-minimum 1 nS.

6. (23) BUS CLK (1) to CB ORIG RDY, CB TERM RDY, CB BUSY. Hold time-minimum 12 nS.

7. (24) DTA TMOUT FF to BUS CLK (1). Setup time-minimum 12 nS.

8. (25) BUS CLK (1) to DTA TMOUT FF. Hold time-minimum 10 nS.

9. (26) BUS CLK (1) to CB GRANT and SUB ORIG RDY. Hold time-minimum 11 nS.

10. (27) MAE LKUP ACT to BUS CLK (1). Setup time-minimum 11 nS.

11. (28) BUS CLK (1) to MAE LKUP ACT. Hold time-minimum 2 nS.

12. (29) FUNCT BIT 2 to BUS CLK (1). Setup time-minimum -5 nS.

13. (30) BUS CLK (1) to FUNCT BIT 2. Hold time-minimum 40 nS.

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TABLE XV. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	3.0 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

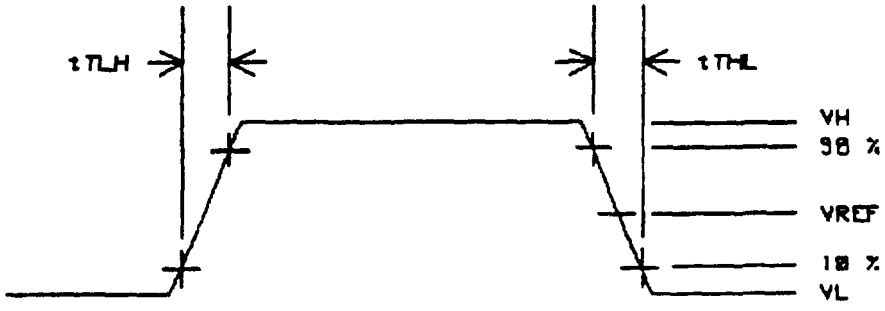
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVI.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XV.

6. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	BUS CLK (1) to CB ADDR TMOUT	-	-	-	42	42	42	nS	12
1	tpd	BUS CLK (1) to CB ATR	-	-	-	42	42	42	nS	12
2	tpd	C4 CLK to IR15 A ENBL	-	-	-	27	27	27	nS	12
3	tpd	BUS CLK (1) to ANY MAIN GNT	-	-	-	48	48	48	nS	12
3	tpd	BUS CLK (1) to ANY MAIN GNT	-	-	-	48	48	48	nS	12
4	tpd	BUS CLK (1) to CB GRANT 3	-	-	-	26	26	26	nS	12
4	tpd	BUS CLK (1) to CB GRANT 2	-	-	-	26	26	26	nS	12
4	tpd	BUS CLK (1) to CB GRANT 1	-	-	-	26	26	26	nS	12
4	tpd	BUS CLK (1) to CB GRANT 0	-	-	-	26	26	26	nS	12
5	tpd	1.6 uS CLK to DTA TMOUT FF	-	-	-	54	54	54	nS	12
6	tpd	BUS CLK (1) to DTA TMOUT	-	-	-	41	41	41	nS	12
7	tpd	EXT CLK to CLK 20 MHZ A	-	-	-	24	24	24	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tpd	EXT CLK to CLK 20 MHZ B	-	-	-	24	24	24	nS	12
7	tpd	EXT CLK to CLK 20 MHZ C	-	-	-	24	24	24	nS	12
7	tpd	EXT CLK to CLK 20 MHZ D	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (1)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (2)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (3)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (4)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (5)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (6)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (7)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (8)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (9)	-	-	-	24	24	24	nS	12
8	tpd	MASTR CB CK to CB CLK (10)	-	-	-	24	24	24	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tpd	BUS CLK (1) to SUB GRANT 0	-	-	-	26	26	26	nS	12
9	tpd	BUS CLK (1) to SUB GRANT 1	-	-	-	26	26	26	nS	12
9	tpd	BUS CLK (1) to SUB GRANT 2	-	-	-	26	26	26	nS	12
9	tpd	BUS CLK (1) to SUB GRANT 3	-	-	-	26	26	26	nS	12
10	tpd	BUS CLK (1) to ANY SUB GNT	-	-	-	48	48	48	nS	12
10	tpd	BUS CLK (1) to ANY GNT 1	-	-	-	48	48	48	nS	12
11	tpd	BUS CLK (1) to CB REQUEST 1	-	-	-	28	28	28	nS	12
12	tpd	MASTR CB CK to CP CLK (1)	-	-	-	25	25	25	nS	12
12	tpd	MASTR CB CK to CP CLK (2)	-	-	-	25	25	25	nS	12
12	tpd	MASTR CB CK to SRC/DEST CK2	-	-	-	25	25	25	nS	12
12	tpd	MASTR CB CK to SRC/DEST CK1	-	-	-	25	25	25	nS	12
12	tpd	MASTR CB CK to SRC A BUS EN	-	-	-	25	25	25	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
12	tpd	CP CLK to CP CLK (1)	-	-	-	25	25	25	nS	12
12	tpd	CP CLK to CP CLK (2)	-	-	-	25	25	25	nS	12
12	tpd	CP CLK to SRC/DEST CK2	-	-	-	25	25	25	nS	12
12	tpd	CP CLK to SRC/DEST CK1	-	-	-	25	25	25	nS	12
12	tpd	CP CLK to SRC A BUS EN	-	-	-	25	25	25	nS	12
13	tpd	SRC/DEST CLK to SRC B BUS EN	-	-	-	44	44	44	nS	12
14	tpd	INS REG 32 to DSABLE ALU 0	-	-	-	19	19	19	nS	12
14	tpd	INS REG 33 to DSABLE ALU 0	-	-	-	19	19	19	nS	12
14	tpd	INS REG 30 to DSABLE ALU 0	-	-	-	19	19	19	nS	12
14	tpd	INS REG 29 to DSABLE ALU 0	-	-	-	19	19	19	nS	12
14	tpd	F3 to DSABLE ALU 0	-	-	-	19	19	19	nS	12
14	tpd	A2 to DSABLE ALU 0	-	-	-	19	19	19	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
15	tpd	BUS CLK (1) to 1.6 uS CLK	-	-	-	35	35	35	nS	12
16	tpd	BUS CLK (1) to CLR MN GNT H	-	-	-	49	49	49	nS	12
17	tpd	BUS CLK (1) to CB ADR TMOUT	-	-	-	33	33	33	nS	12
17	tpd	BUS CLK (1) to CB ATR	-	-	-	33	33	33	nS	12
18	tsu	CB SUB REQ 1 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB SUB REQ 2 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB SUB REQ 3 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB REQUEST 0 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB REQUEST 1 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB REQUEST 2 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB REQUEST 3 to BUS CLK (1)	13	13	13	-	-	-	nS	12
18	tsu	CB SUB REQ 0 to BUS CLK (1)	13	13	13	-	-	-	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
19	th	BUS CLK (1) to CB ATR	11	11	11	-	-	-	nS	12
20	tsu	CB WR EN to BUS CLK (1)	24	24	24	-	-	-	nS	12
21	th	BUS CLK (1) to CB WR EN	8	8	8	-	-	-	nS	12
22	tsu	CB ORIG RDY to BUS CLK (1)	1	1	1	-	-	-	nS	12
22	tsu	DTA TERM RDY to BUS CLK (1)	1	1	1	-	-	-	nS	12
22	tsu	CB BUSY to BUS CLK (1)	1	1	1	-	-	-	nS	12
23	th	BUS CLK (1) to CB ORIG RDY	12	12	12	-	-	-	nS	12
23	th	BUS CLK (1) to DTA TERM RDY	12	12	12	-	-	-	nS	12
23	th	BUS CLK (1) to CB BUSY	12	12	12	-	-	-	nS	12
24	tsu	DTA TMOUT FF to BUS CLK (1)	12	12	12	-	-	-	nS	12
25	th	BUS CLK (1) to DTA TMOUT FF	10	10	10	-	-	-	nS	12
26	th	BUS CLK (1) to CB GRANT	11	11	11	-	-	-	nS	12
26	th	BUS CLK (1) to SUB ORIG RDY	11	11	11	-	-	-	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
27	tsu	MAE LKUP ACT to BUS CLK (1)	11	11	11	-	-	-	nS	12
28	th	BUS CLK (1) to MAE LKUP ACT	2	2	2	-	-	-	nS	12
29	tsu	FUNCT BIT 2 to BUS CLK (1)	-5	-5	-5	-	-	-	nS	12
30	th	BUS CLK (1) to FUNCT BIT 2	40	40	40	-	-	-	nS	12

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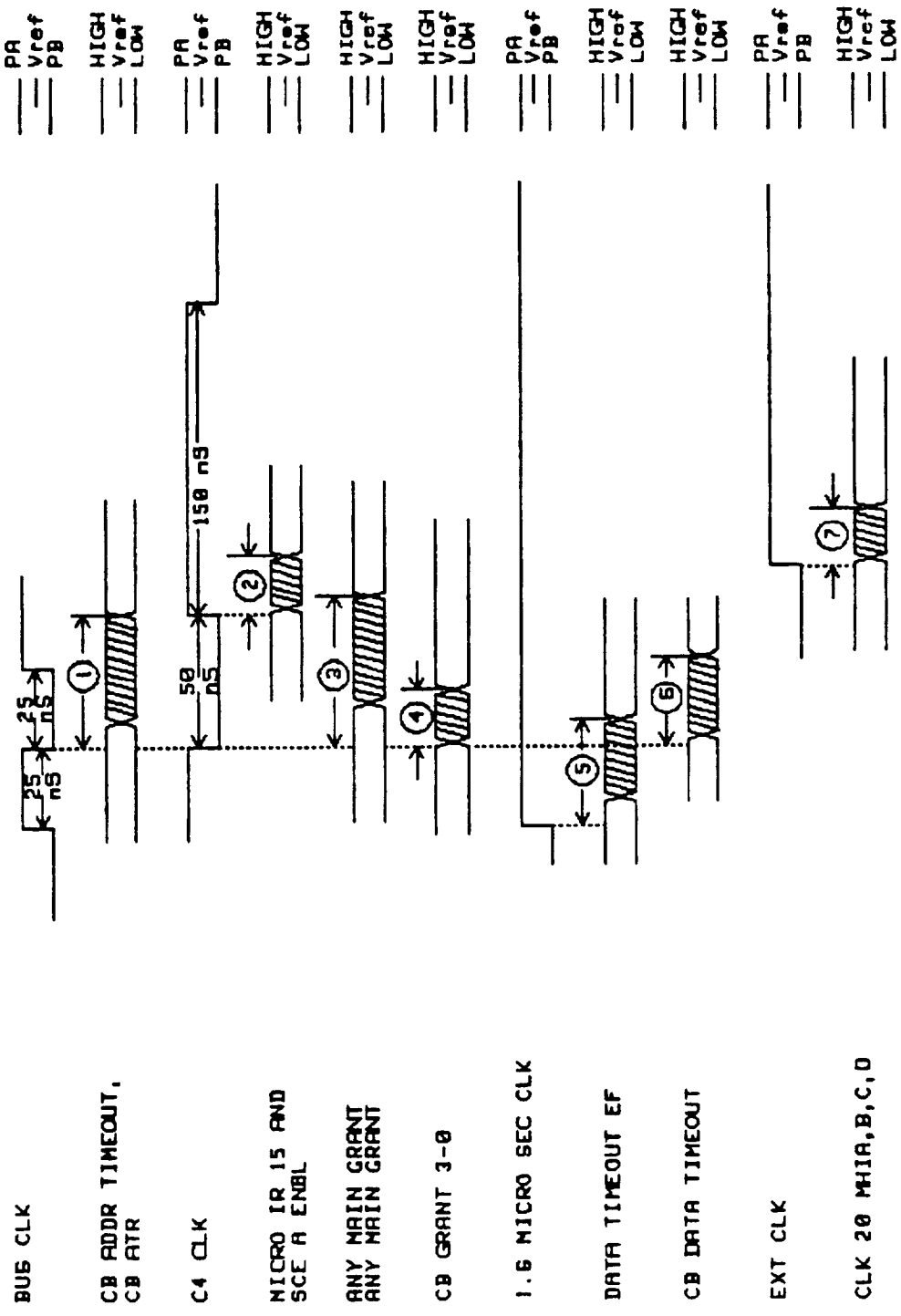


FIGURE 12a. Dynamic waveforms for table XVI.

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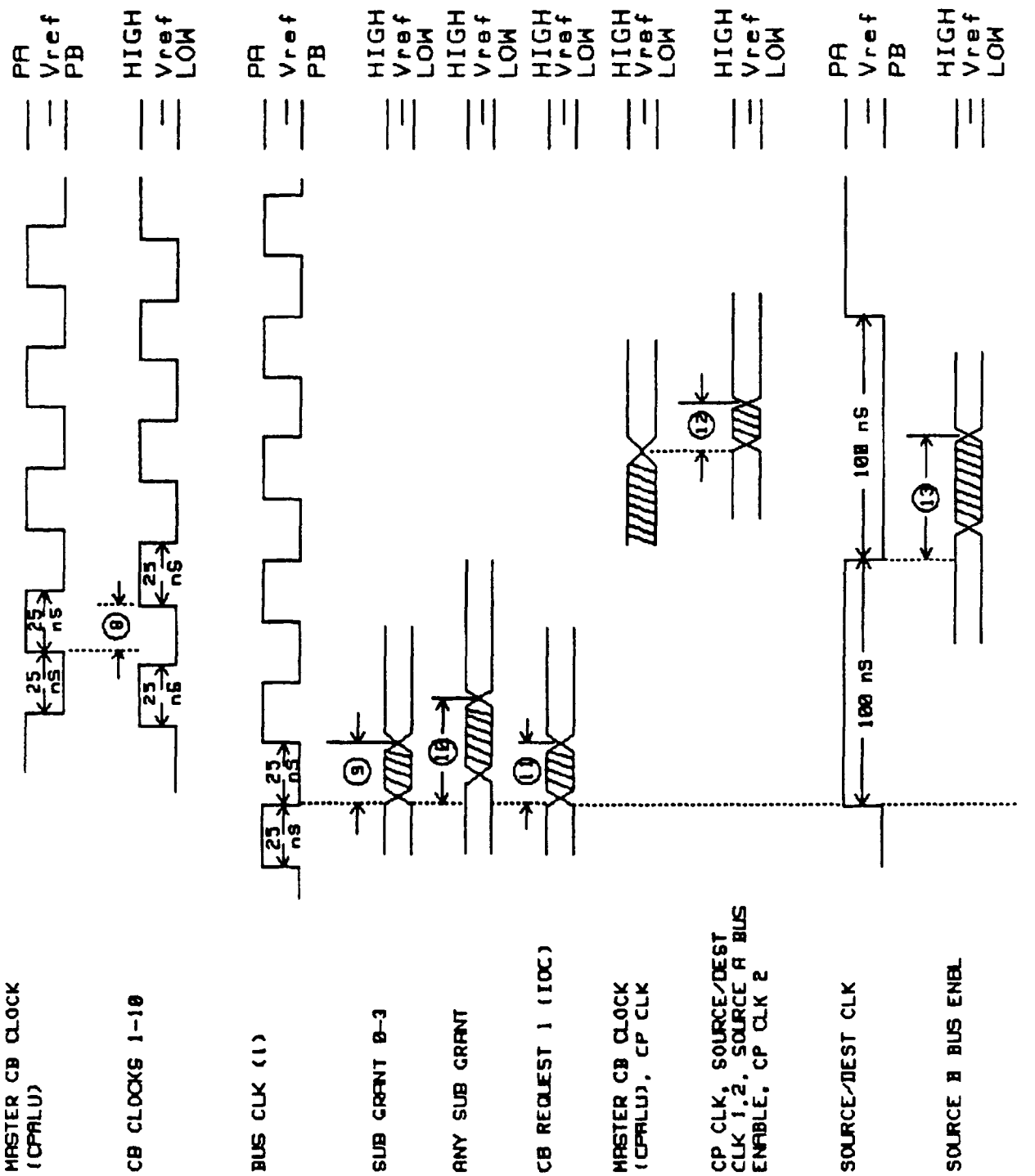


FIGURE 12b. Dynamic waveforms for table XVI.

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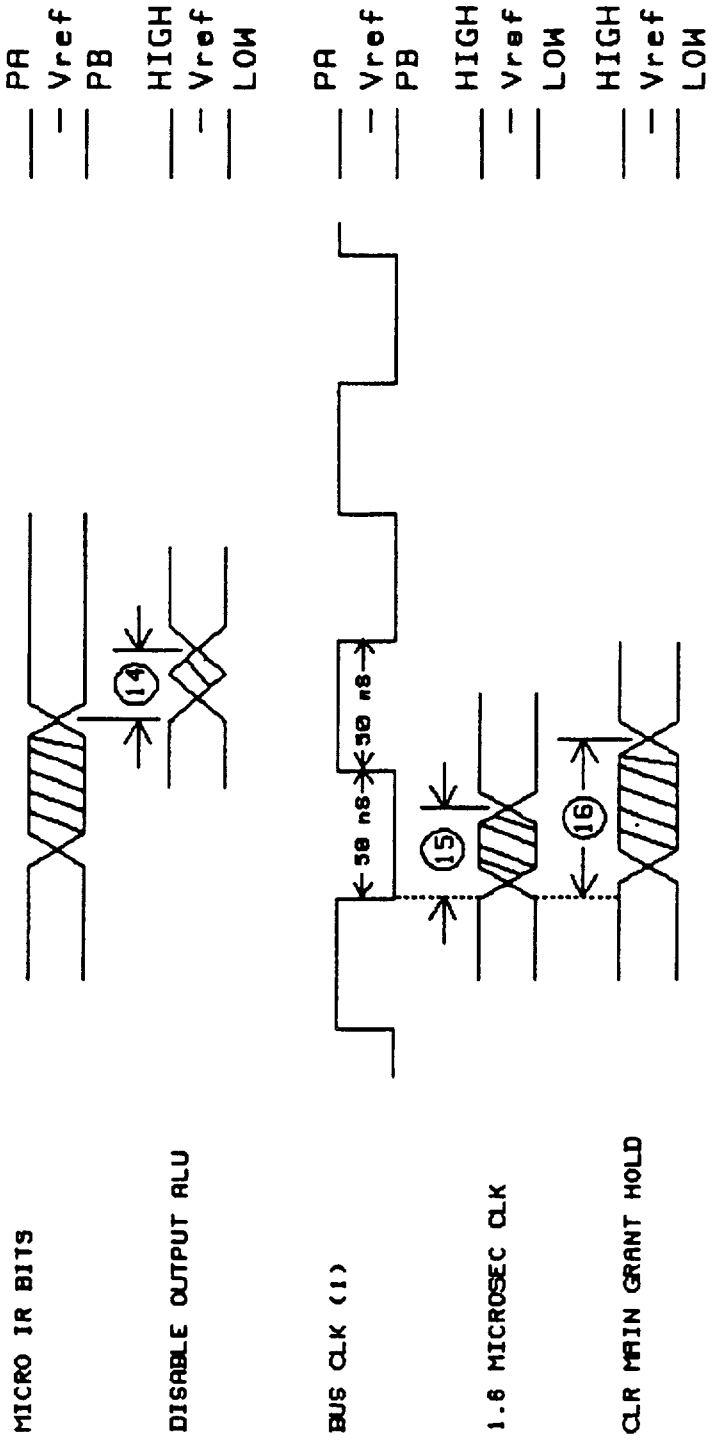


FIGURE 12c. Dynamic waveforms for table XVI.

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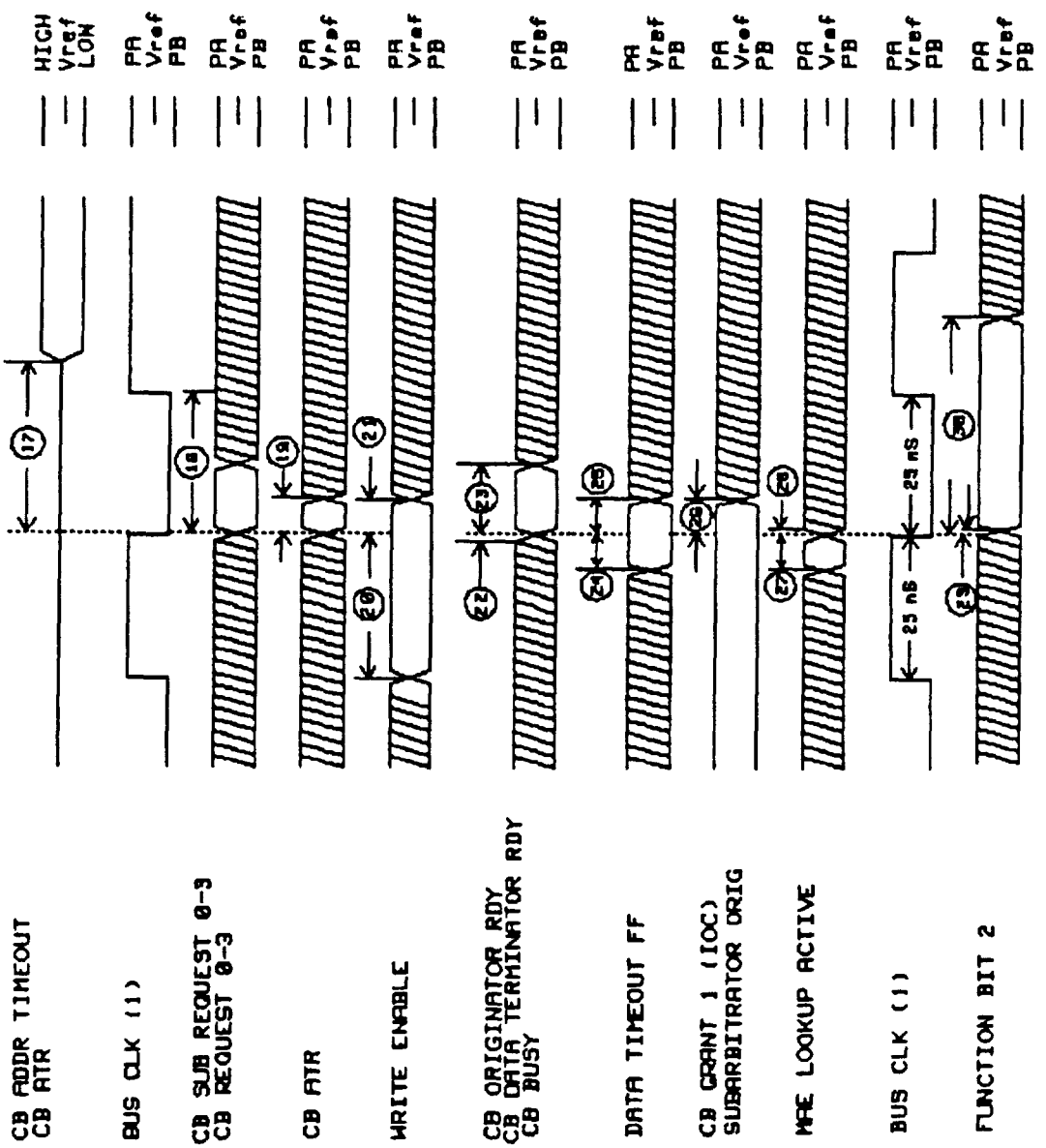


FIGURE 12d. Dynamic waveforms for table XVI.

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TABLE XVII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	CB SUB REQ 1	2	4	100	pF	1-4
R2	CB SUB REQ 0	3				
R3	CB REQUEST 3	4				
R4	CB REQUEST 2	5				
R5	CB REQUEST 1	6	4	100	pF	1-4
R6	CB REQUEST 0	7	4	75	pF	1-4
R7	INS REG 32	14	2	50	pF	1-4
R8	INS REG 33	15				
R9	INS REG 30	16				
R10	INS REG 29	17	2	50	pF	1-4
R11	DTA TMOUT FF	18	2	125	pF	1-4
R12	EN EXT CLK	19	6	75	pF	1-4
R13	EXT CLK	20	2	50	pF	1-4
R14	MICRO IR 15	29	2	50	pF	1-4
R15	EN IC TMOUT	30	4	50	pF	1-4
R16	1.6 uS CLK	42	4	75	pF	1-4
R17	CB SUB REQ 2	44	4	100	pF	1-4
R18	CB SUB REQ 3	45	4	125	pF	1-4
R19	CP CLK	54	2	75	pF	1-4
R20	MAINT PRCSR	58	4	75	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	F3	59	2	50	pF	1-4
R22	MASTR CB CK	61	2	75	pF	1-4
R23	A2	65	2	50	pF	1-4
R24	C4 CLK	66	4	75	pF	1-4
R25	SRC/DEST CLK	67	2	50	pF	1-4
R26	FUNCT BIT 2	74	2	75	pF	1-4
R27	BUS CLK (1)	76	4	75	pF	1-4
R28	CB ORIG RDY	85				
R29	DTA TERM RDY	86	4	75	pF	1-4
R30	CB MA CLR	87	2	50	pF	1-4
R31	CB BUSY	88	4	75	pF	1-4
R32	CB GRANT	91	2	50	pF	1-4
R33	MAE LKUP ACT	92	4	100	pF	1-4
R34	SUB ORG RDY	93	4	75	pF	1-4
R35	CB WR EN	95	4	100	pF	1-4
Outputs						
R36	CB GRANT 3	8	4	75	pF	1-4
R37	CB GRANT 2	9				
R38	CB GRANT 1	11				
R39	CB GRANT 0	12	4	75	pF	1-4
R40	ANY MAIN GNT	13	10	150	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	19HZ ICT B	21	2	50	pF	1-4
R42	CLK 20 MHZ B	22				
R43	CLK 20 MHZ D	23				
R44	19HZ DSTK CK	24				
R45	CLK 20 MHZ C	26				
R46	19HZ ICT A	27				
R47	ANY MAIN GNT	28				
R48	IR15 A ENBL	32				
R49	CB CLK (4)	33				
R50	CB CLK (5)	34				
R51	CB CLK (6)	35				
R52	CB CLK (7)	36				
R53	CB CLK (8)	37				
R54	CB CLK (9)	38				
R55	CB CLK (10)	39	2	50	pF	1-4
R56	25.6 uS CLK	43	4	75	pF	1-4
R57	CLK 20 MHZ A	46	2	100	pF	1-4
R58	CLR MN GNT H	47	4	175	pF	1-4
R59	ACTIV PULLUP	48	2	50	pF	1-4
R60	CB DTA TMOUT	49	4	150	pF	1-4
R61	CB ADR TMOUT	50	4	75	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R62	CB CLK (3)	53	2	75	pF	1-4
R63	1.6 uS CLK	56	2	50	pF	1-4
R64	SRC B BUS EN	63				
R65	CP CLK (1)	69				
R66	CB CLK (2)	72				
R67	CB CLK (1)	73				
R68	SRC A BUS EN	80	2	50	pF	1-4
R69	DSABLE ALU O	82	4	75	pF	1-4
R70	CP CLK (2)	83	2	50	pF	1-4
R71	CB REQUEST 1	89				
R72	ANY SUB GNT	94	2	50	pF	1-4
R73	SUB GRANT 0	96	4	75	pF	1-4
R74	SUB GRANT 1	97				
R75	SUB GRANT 2	98	4	75	pF	1-4
R76	SUB GRANT 3	99	8	100	pF	1-4
R77	ANY GNT 1	100	4	75	pF	1-4
R78	SRC/DEST CK2	77				
R79	SRC/DEST CK1	79	4	75	pF	1-4
R80	CB ATR	84	6	125	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R80
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R80
Capacitance measurement	NA	± 4.8 pF	R1-R5
	NA	± 3.55 pF	R6
	NA	± 2.4 pF	R7-R10
	NA	± 6.15 pF	R11
	NA	± 3.45 pF	R12
	NA	± 2.4 pF	R13,R14
	NA	± 2.3 pF	R15
	NA	± 3.55 pF	R16
	NA	± 4.8 pF	R17
	NA	± 6.05 pF	R18
	NA	± 3.65 pF	R19
	NA	± 3.55 pF	R20
	NA	± 2.4 pF	R21
	NA	± 4.9 pF	R22
	NA	± 2.4 pF	R23
	NA	± 3.55 pF	R24
	NA	± 2.4 pF	R25
	NA	± 3.65 pF	R26
	NA	± 3.55 pF	R27-R29
	NA	± 2.4 pF	R30
	NA	± 3.55 pF	R31
	NA	± 2.4 pF	R32
	NA	± 4.8 pF	R33
	NA	± 3.55 pF	R34
	NA	± 4.8 pF	R35
	NA	± 3.55 pF	R36-R39
	NA	± 7.0 pF	R40
	NA	± 2.4 pF	R41-R55
	NA	± 3.55 pF	R56
	NA	± 4.9 pF	R57
	NA	± 8.55 pF	R58

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 2.4 pF	R59
	NA	± 7.3 pF	R60
	NA	± 3.55 pF	R61
	NA	± 3.65 pF	R62
	NA	± 2.4 pF	R63-R68
	NA	± 3.55 pF	R69
	NA	± 2.4 pF	R70-R72
	NA	± 3.55 pF	R73-R75
	NA	± 4.6 pF	R76
	NA	± 3.55 pF	R77-R79
	NA	± 5.95 pF	R80

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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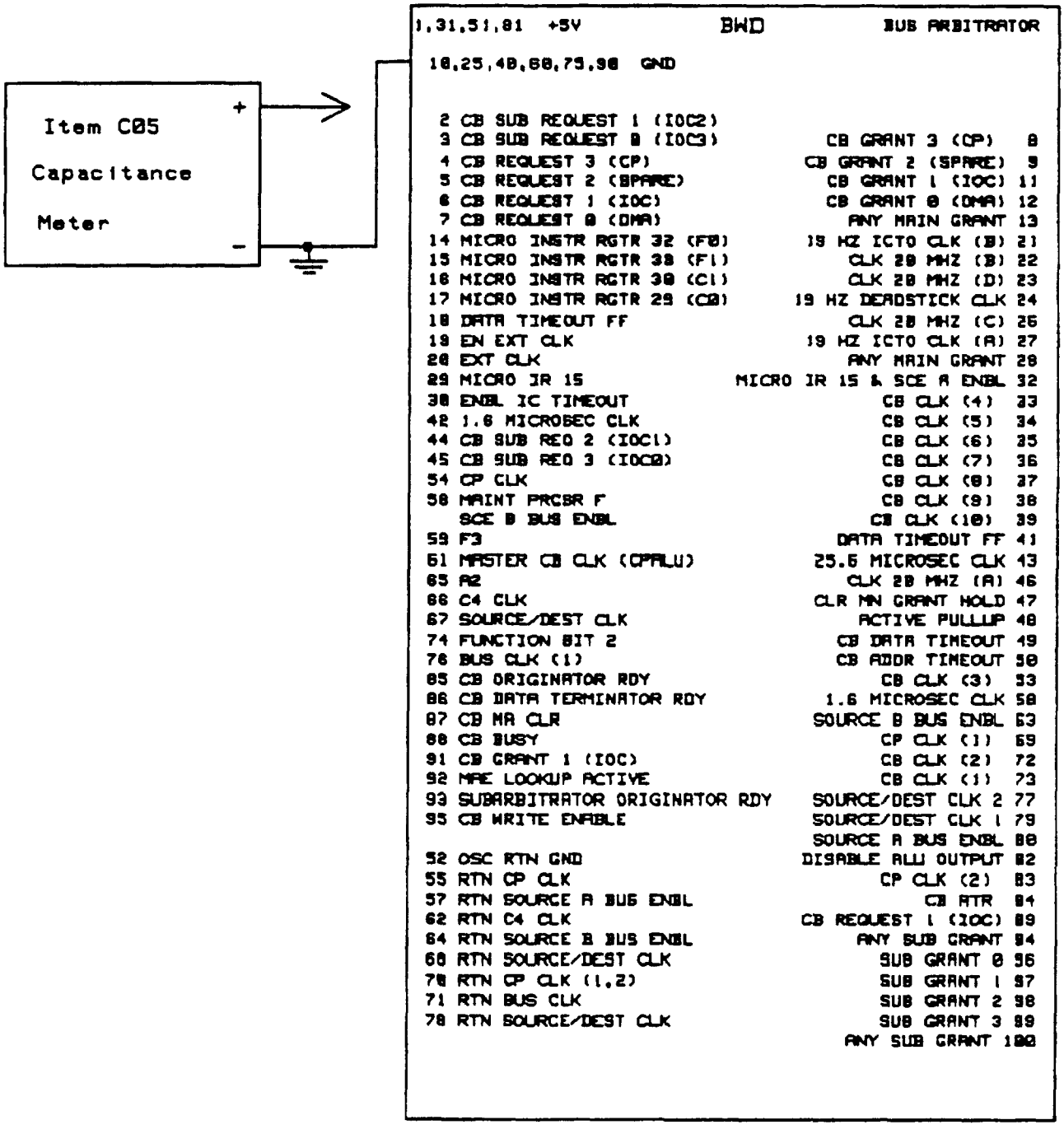


FIGURE 13. Input and output capacitance test circuit for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1	0	400	mV	1-4	14,15
T2	PSTA	51	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	3.0 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVIII.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XVIII.

(d) Using the oscilloscope, D06, inspect for the most negative spike as indicated on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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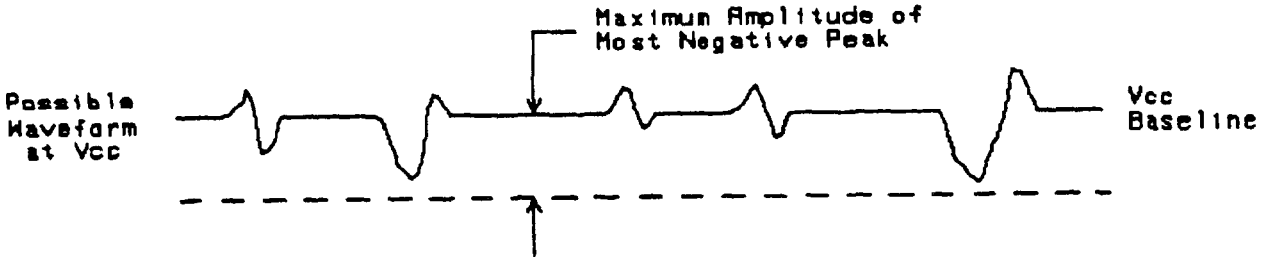
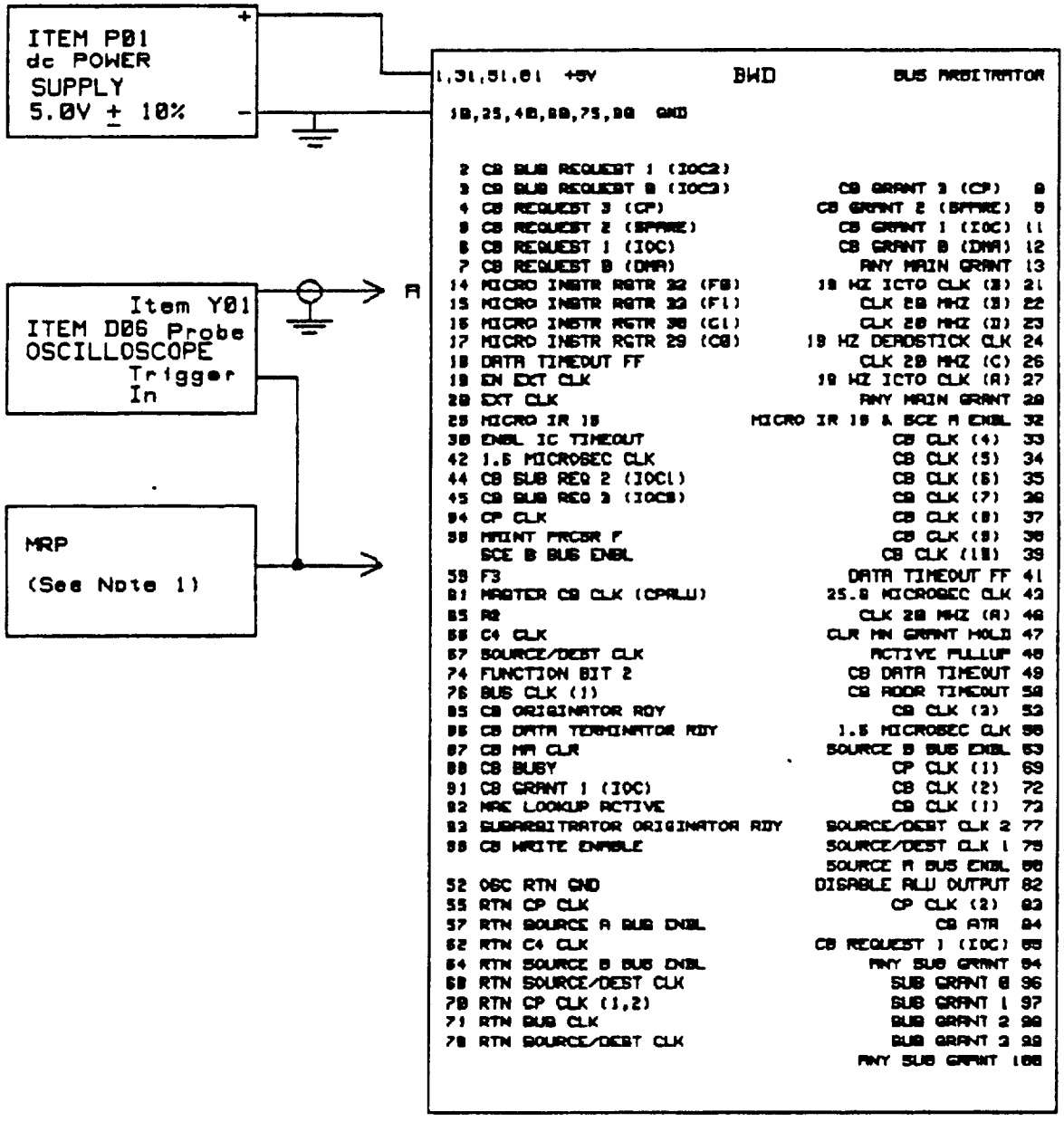


FIGURE 14. Power supply transient amplitude waveform for table XVIII.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XVIII.

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